Advanced Biotechnology

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Professor of Bionanoscience
CEBT II - Michaelmas 2025



UNIVERSITY OF CAMBRIDGE

Lecture 16

Bonus Track:

Engineering synthetic biomolecular condensates and organelles

(mainly with nucleic acids)



UNIVERSITY OF CAMBRIDGE

Lecture outline

1) Why build synthetic biomolecular condensates?
2) Engineering synthetic condensates with DNA
3) Engineering synthetic condensates with RNA
4) Expressing synthetic RNA condensates in cells

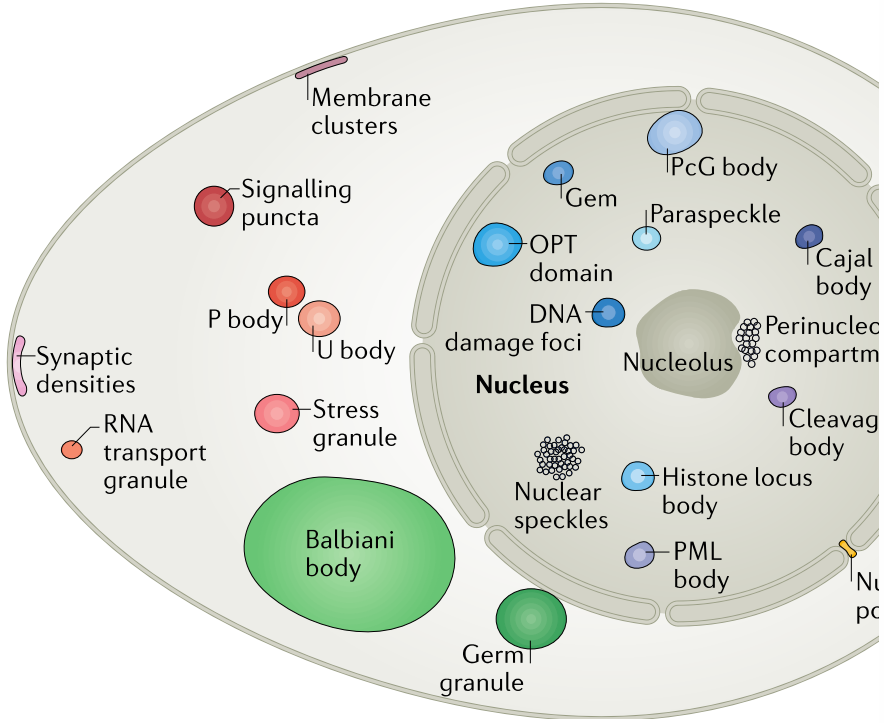
Learning outcomes

- In this lecture we will apply many of the techniques and concepts we studied in the course to address research questions
- You should aim to recognize these methods and concepts and understand why and how they are applied
- There will be quizzes, so pay attention!

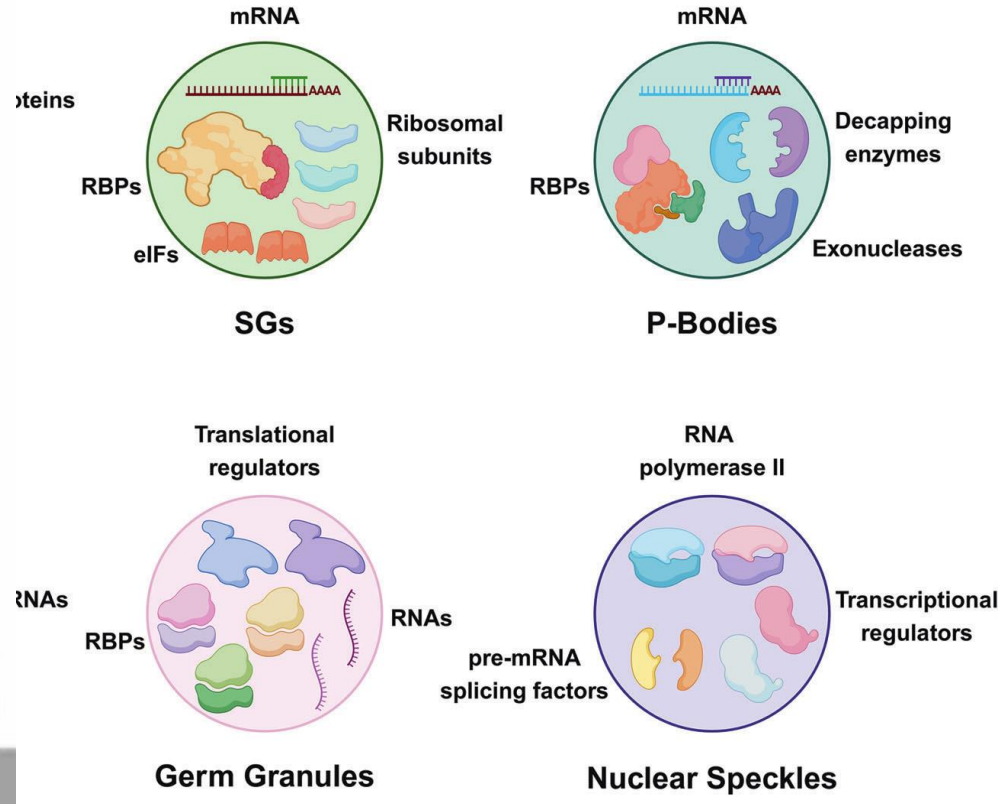
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Biomolecular condensates and membrane-less organelles





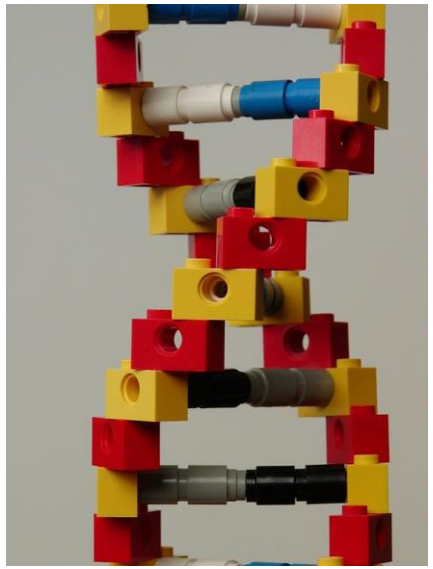


Nat. Rev. Mol. Cell Biol., 18, 285–298 (2017)

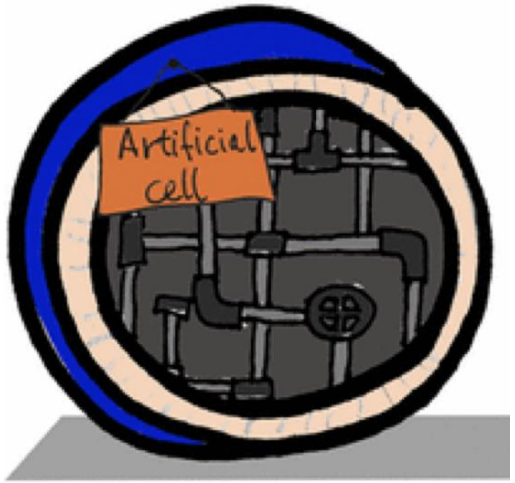
Signal Transduction and Targeted Therapy 9:305 (2024)

Model and synthetic condensates

Learning about biology by "building new biology"



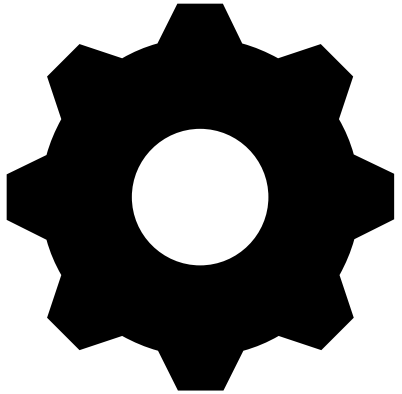
Synthetic organelles for synthetic (and biological) cells



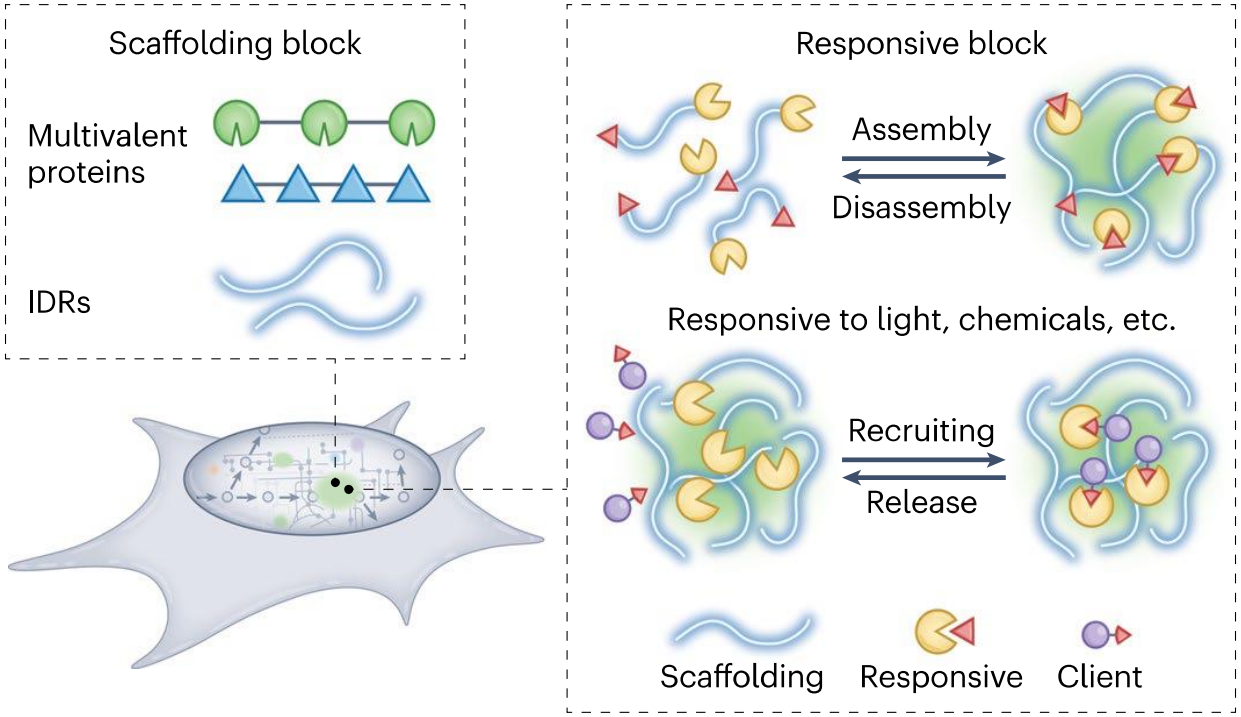
Acc. Chem. Res., 50 769–777, 2017

Applications

- Biocatalysis / biomanufacturing
- Drug screening
- Drug delivery
- …



Protein and peptide designs



Nat. Chem. Bio., 1330–1340, 2022

Advantages

- Genetically encoded
- Built in functions and stimuli responsiveness
- ML protein design tool

Disadvantages

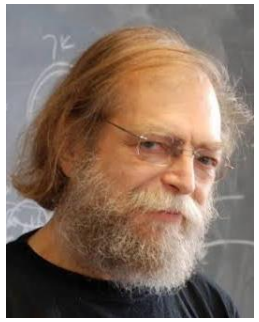
- Rational design difficult
- Interactions complex and non-specific
- Difficult to predict (esp. for IDRs)

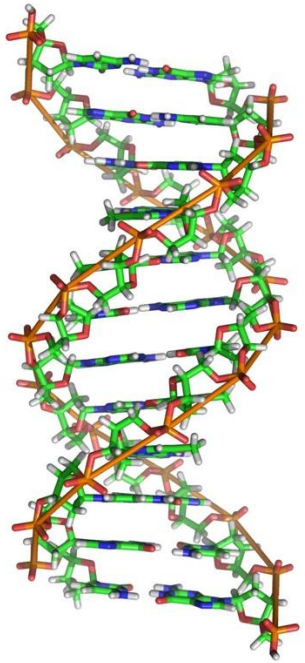
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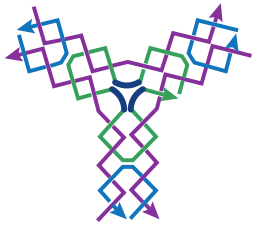
DNA nanotechnology

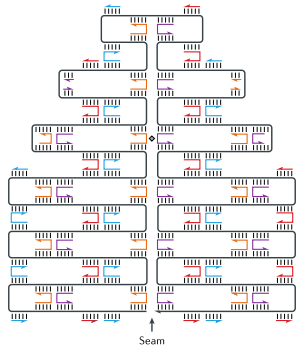
Ned Seeman



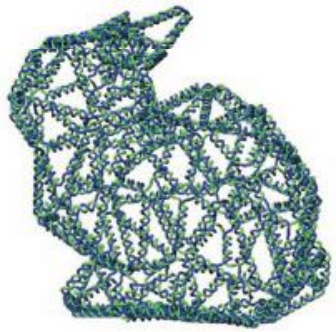






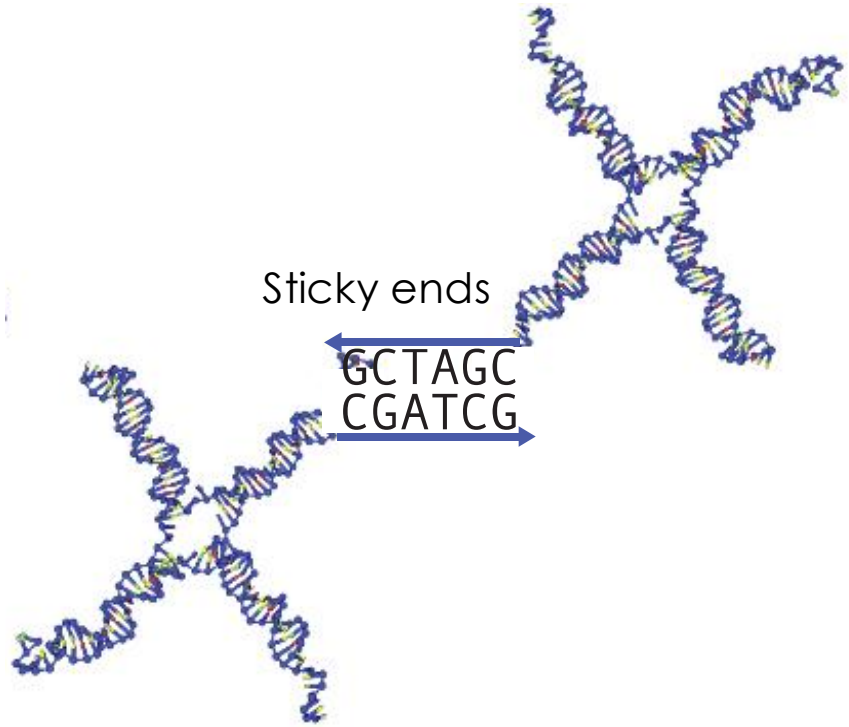


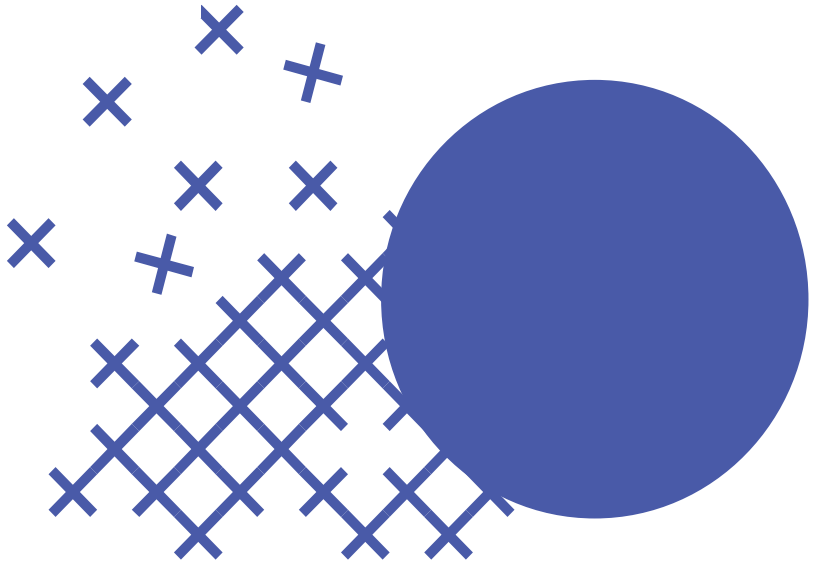


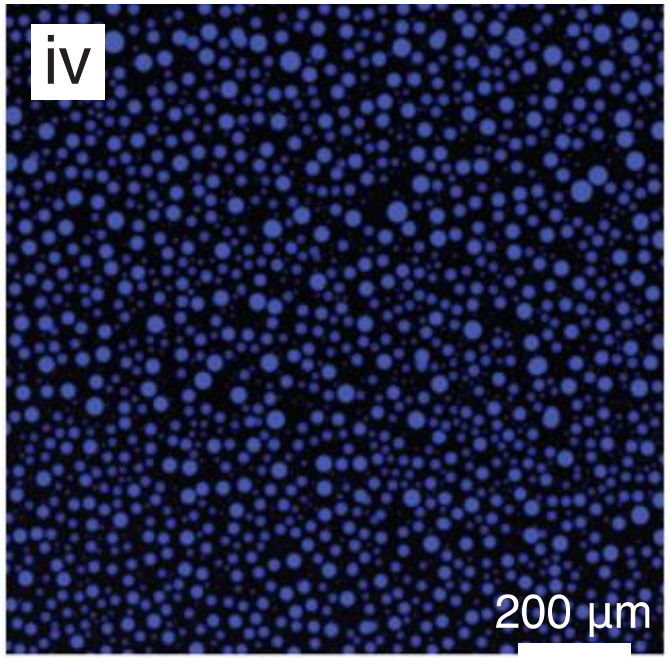


Synthetic DNA condensates

DNA Nanostars





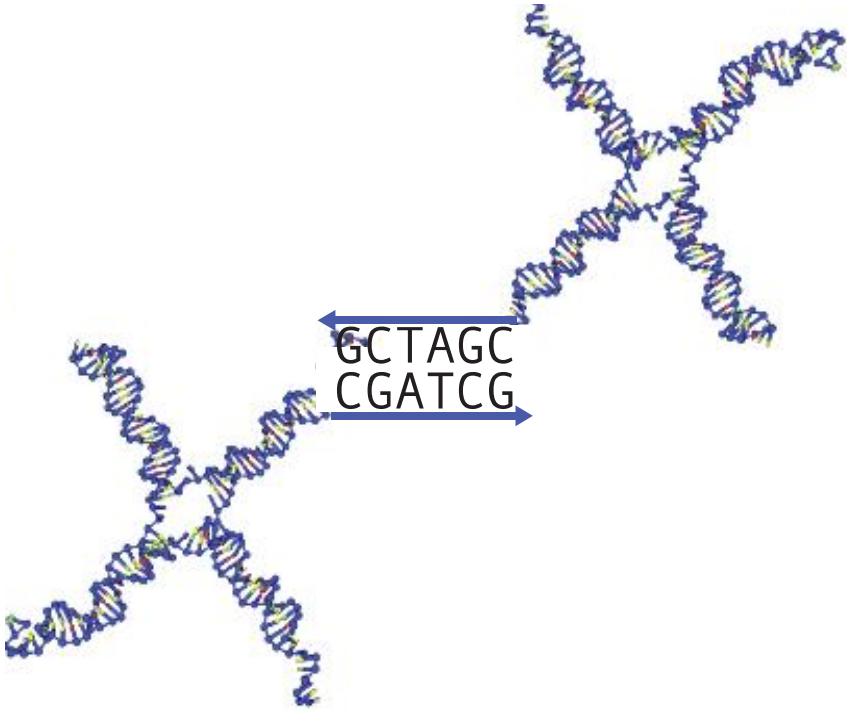


Rep. Prog. Phys. 87, 066601 (2024)

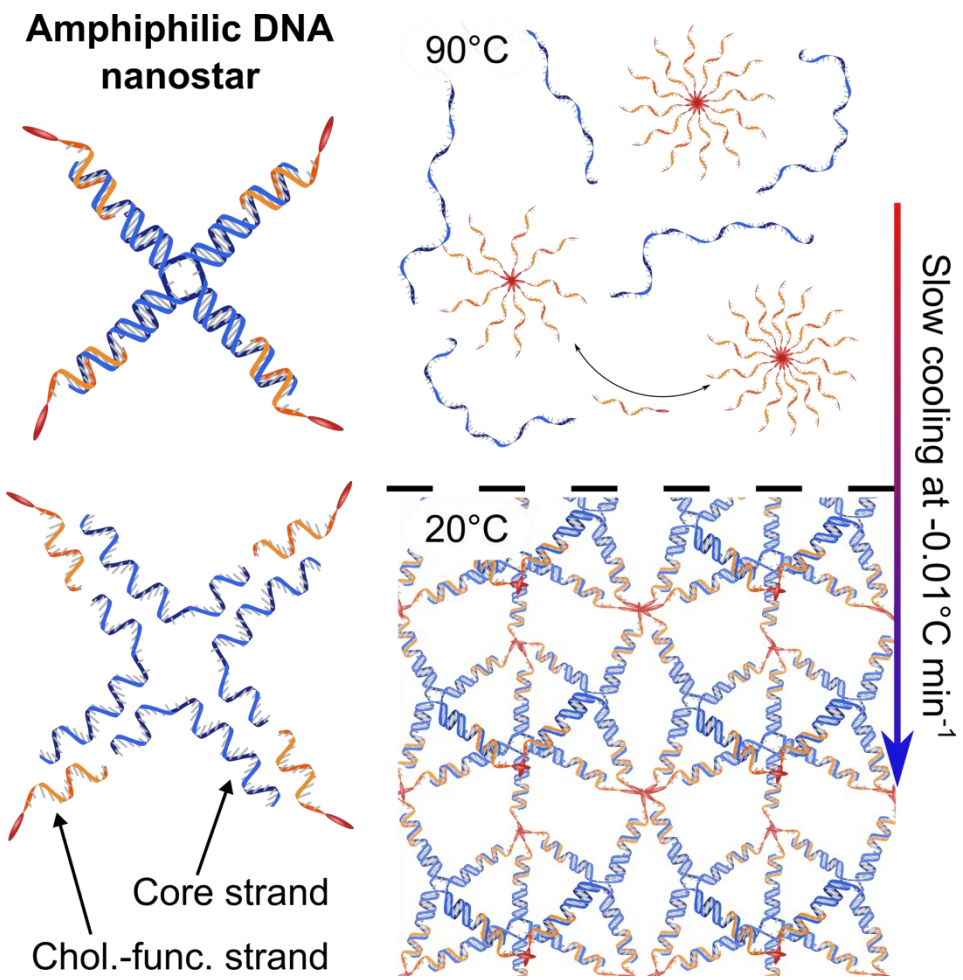
Quiz 1

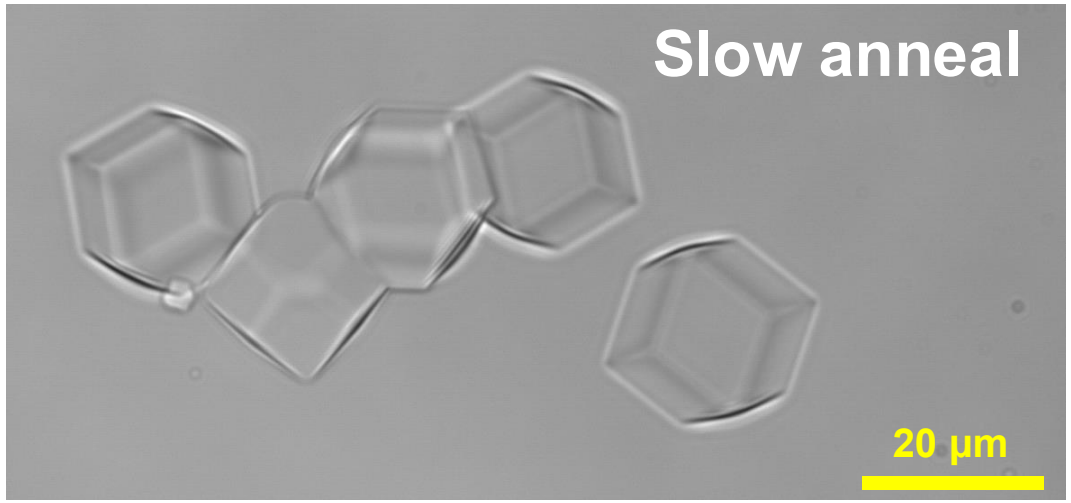
My DNA nanostars have sticky ends of sequence 5' CGATCG 3', but I found that the melting temperature of the condensates is too low. Which of the following sticky end sequences is my best bet to increase the melting temperature?

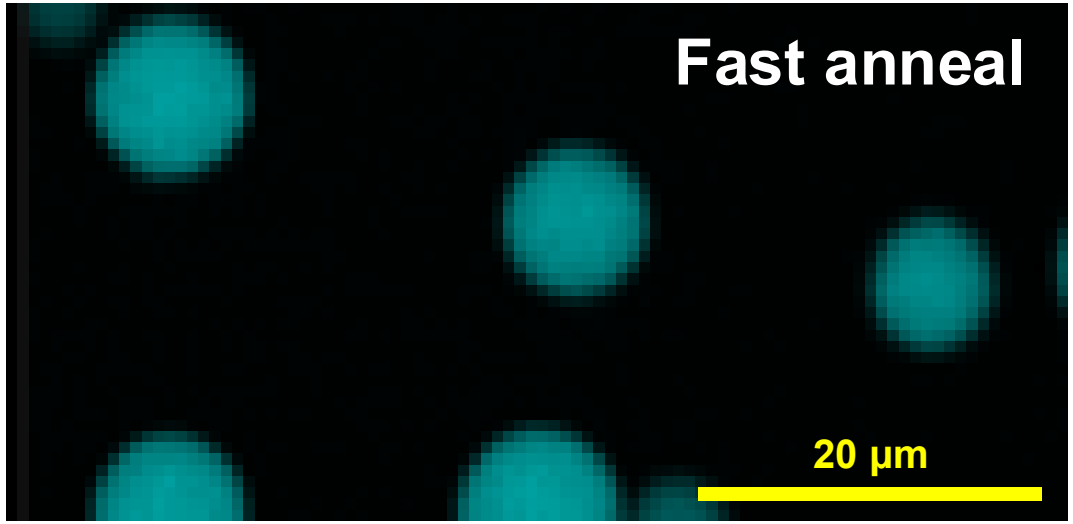
1) 5' GGGGCCCC 3'
2) 5' CAGATCTG 3'
3) 5' CCGATGCG 3'
4) 5' CGCG 3'



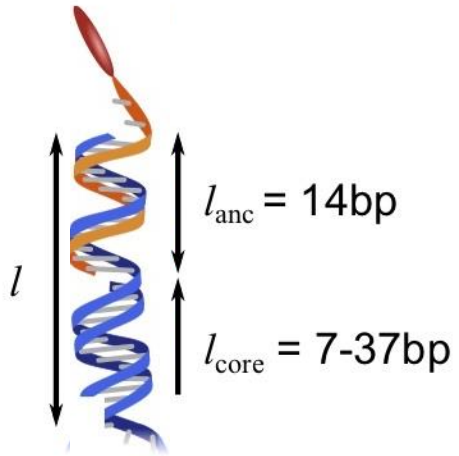
Amphiphilic DNA phases

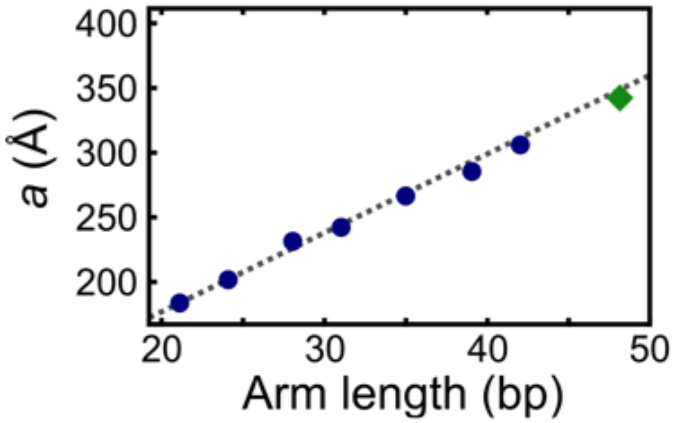


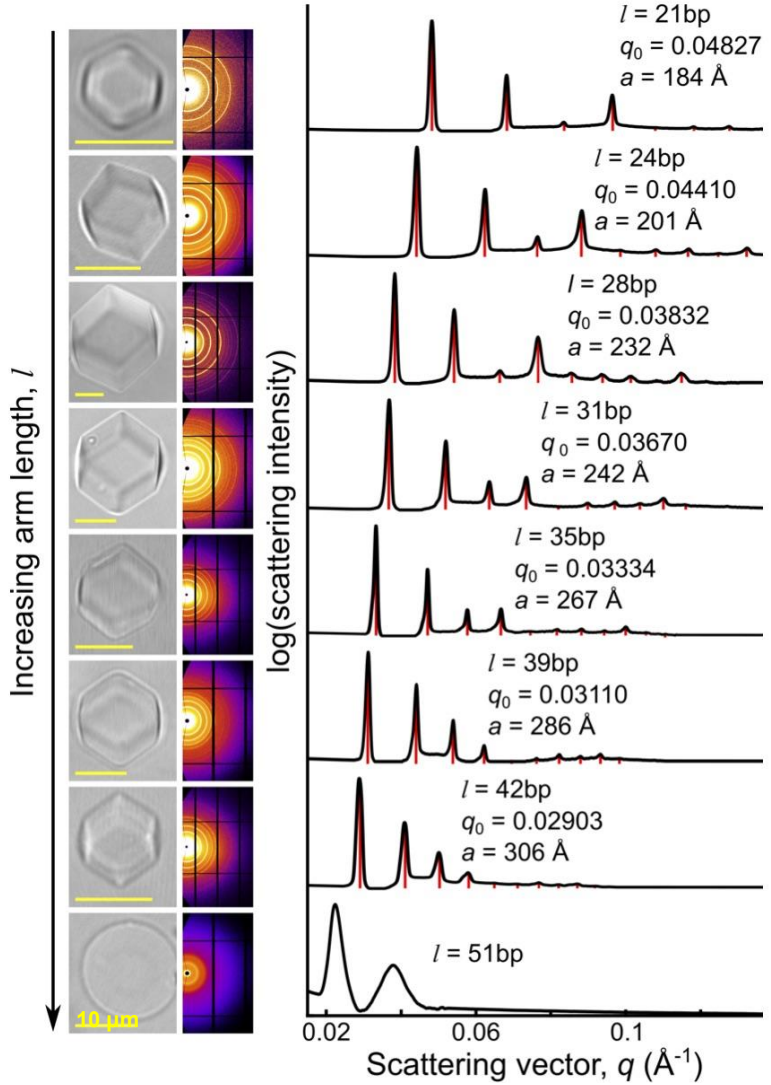


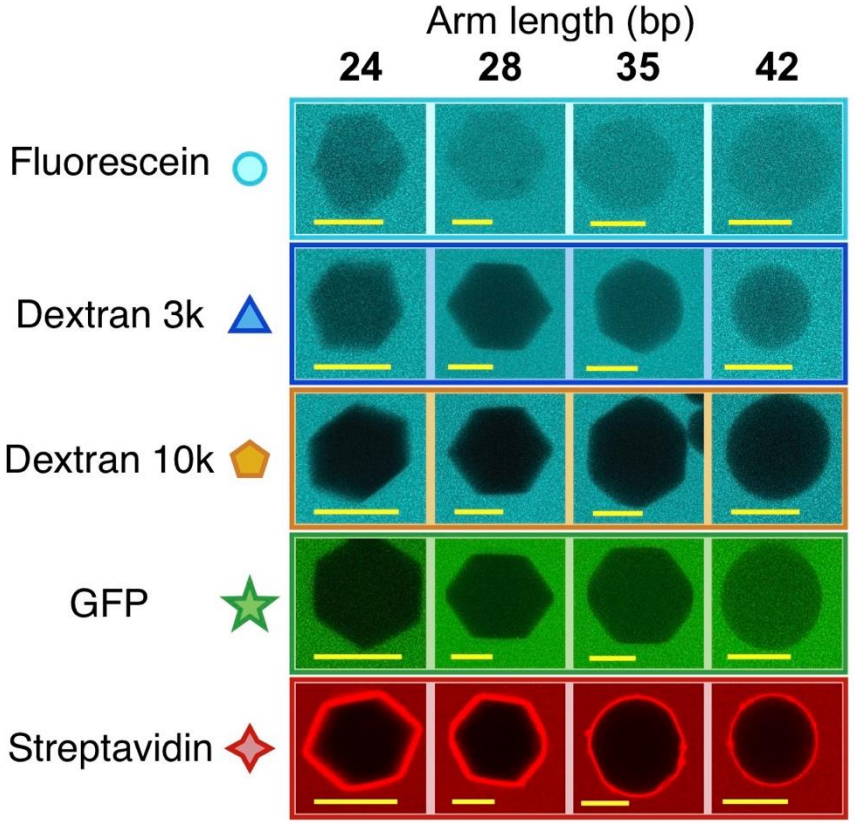


Controlling pore size and diffusive transport









Nano Lett., 17(5) 3276 (2017), *JACS* 140(45)15384 (2018)

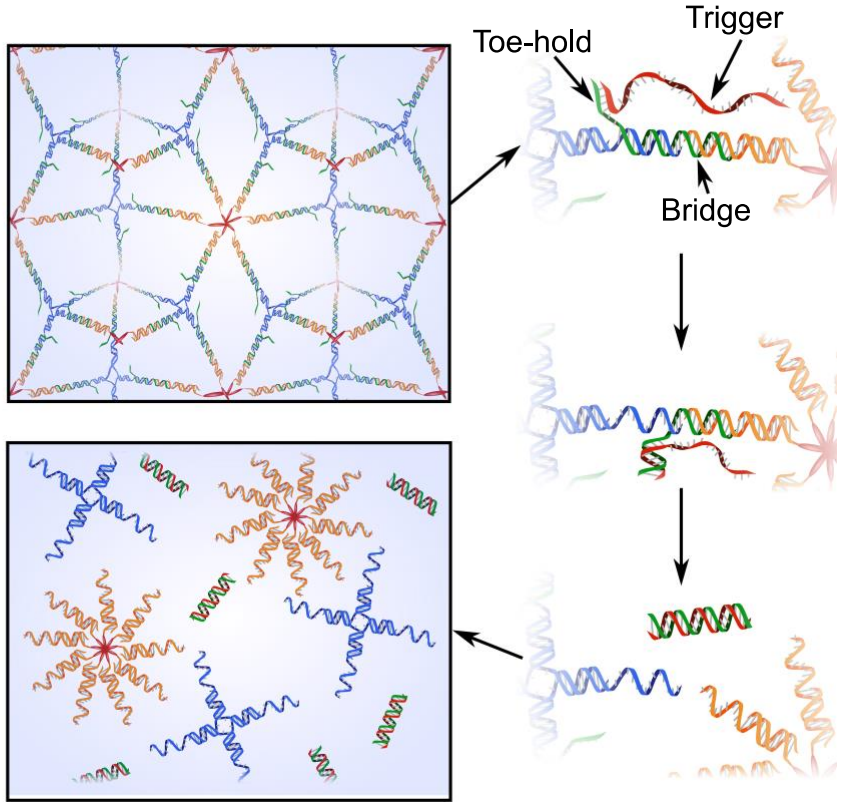
Nano Lett., 22, 602–611, (2022), *Nature Comm.*, 12:4743 (2021)

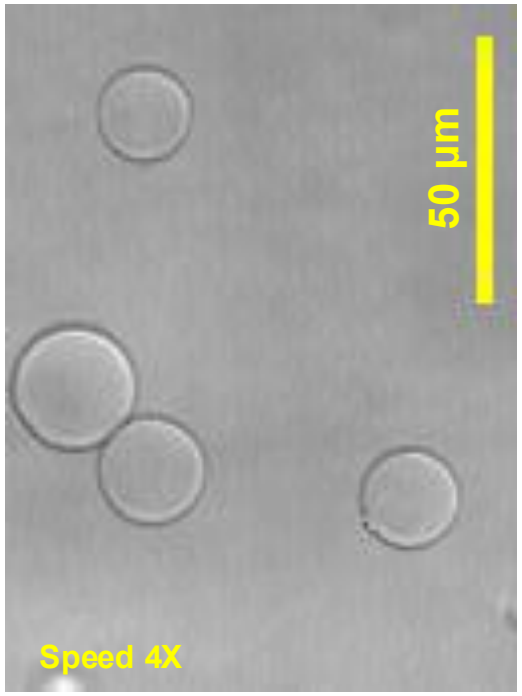
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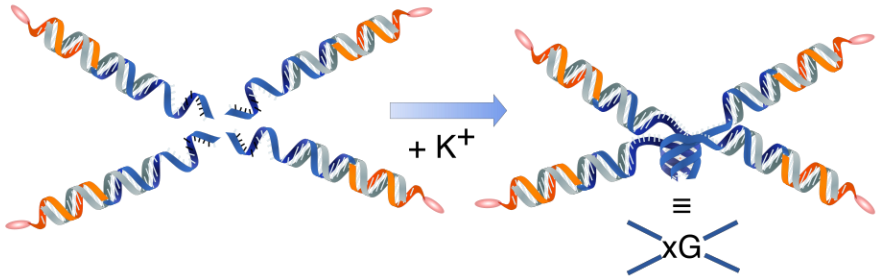
Structural responsiveness

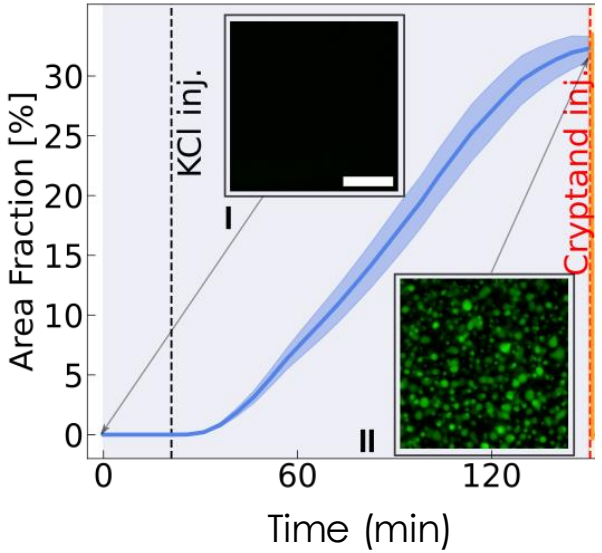
Strand displacement





Cations (G-quadruplexes)

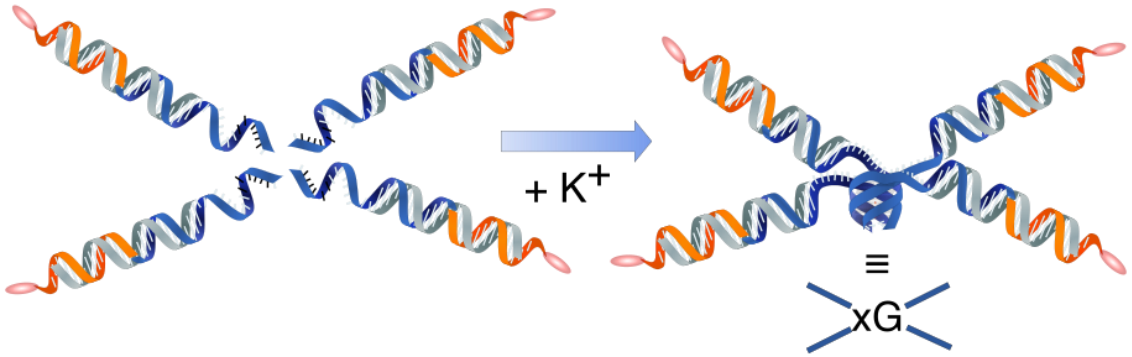




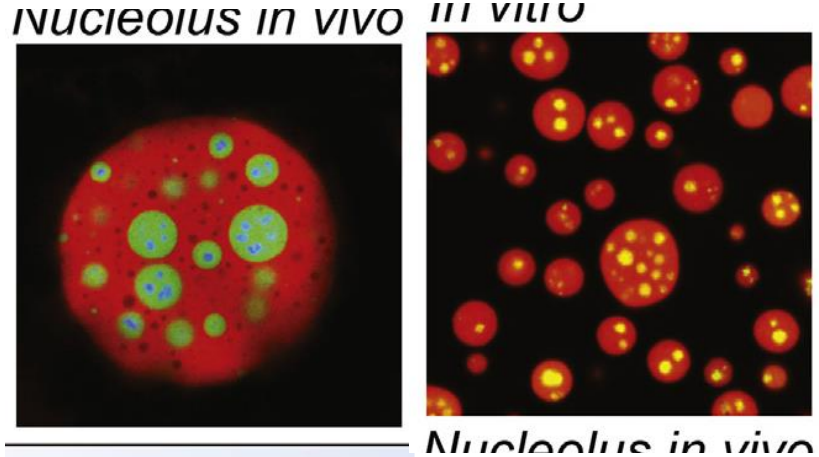
Quiz 2

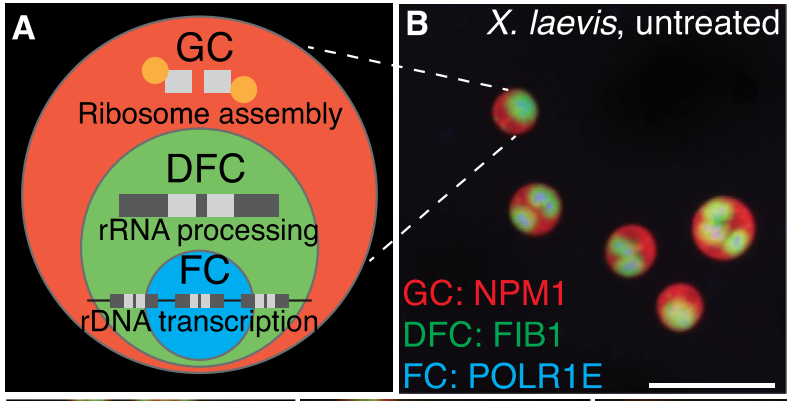
I want to build DNA nanostructures that are pH, rather than cation-responsive. Which DNA structures can be useful for this?

1) A-type double helices
2) i-motifs
3) DNA triplexes
4) Some of the above
5) All of the above

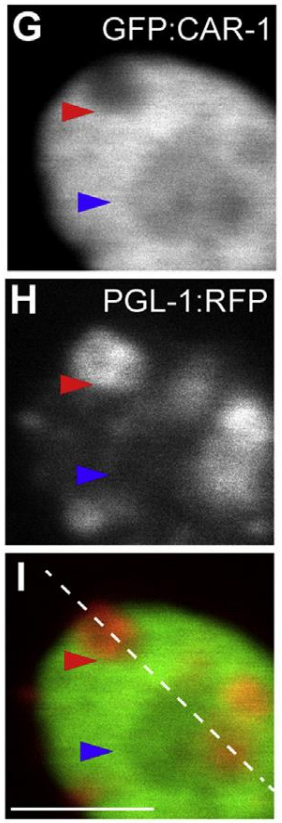


Multiphase condensates



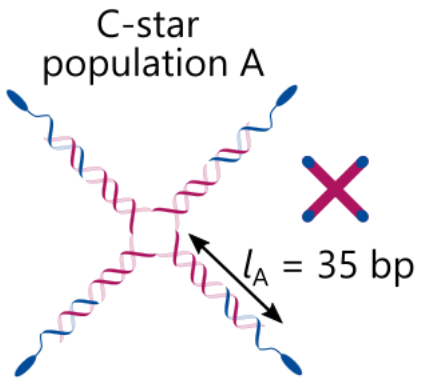


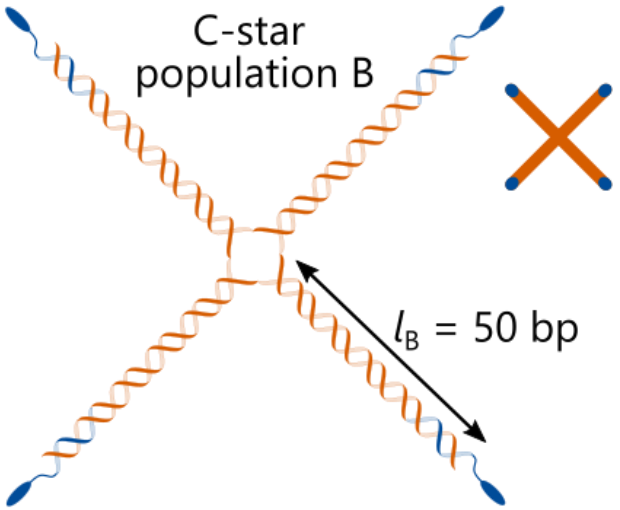
Feric *et al.*, Cell, 165, 1686, 2016

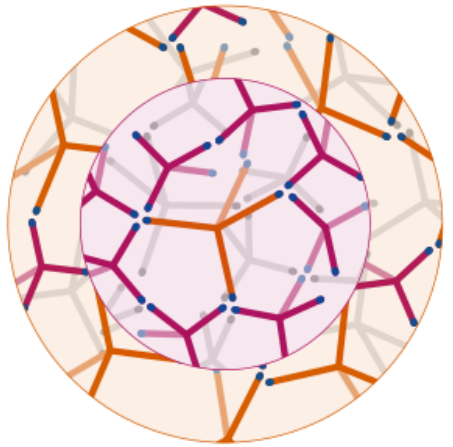


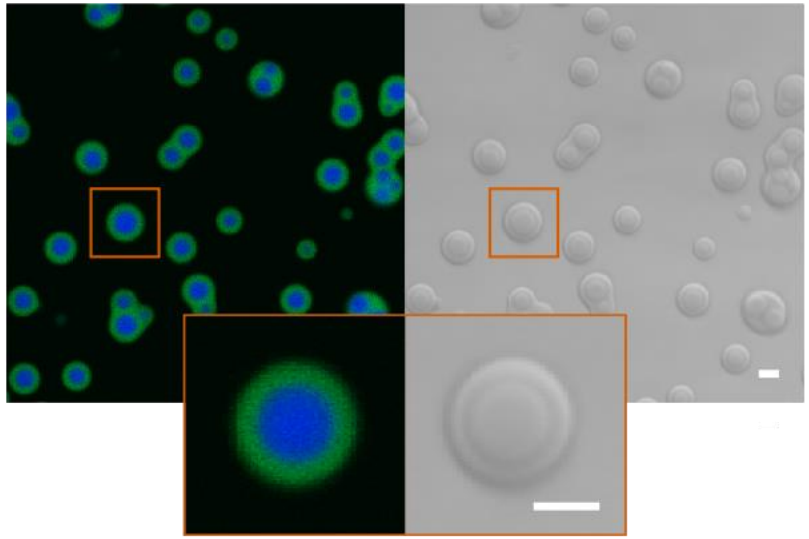
Dev. Cell 27, 161, 28, 2013

Patterning with size-induced phase separation



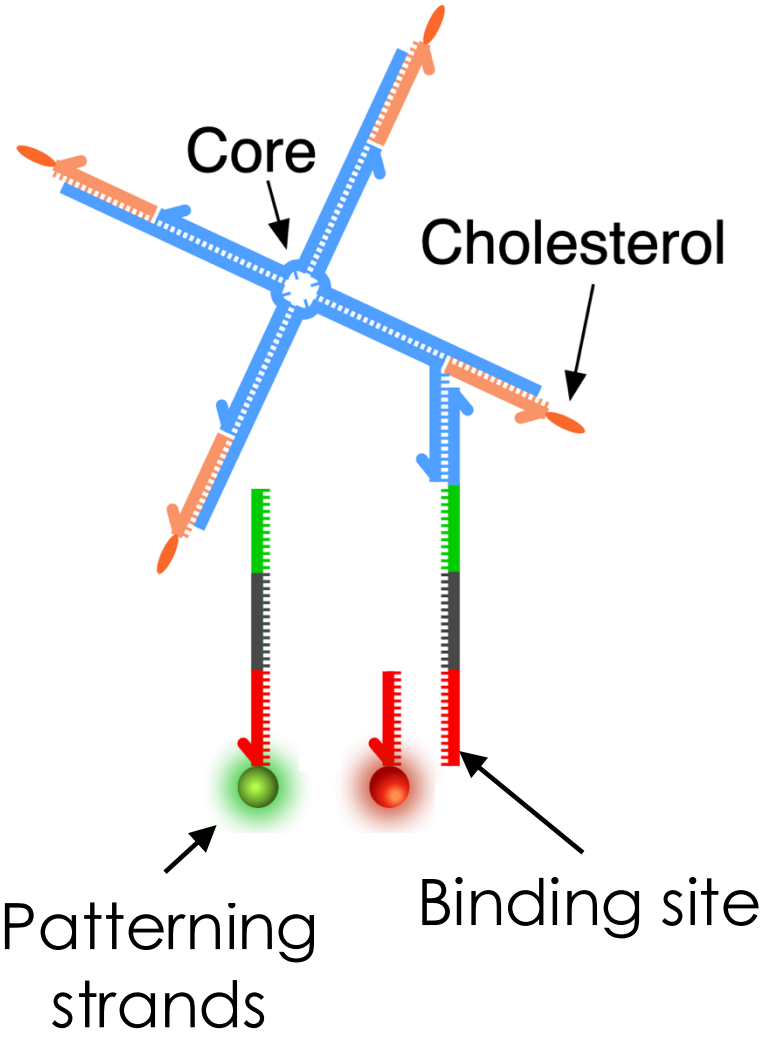






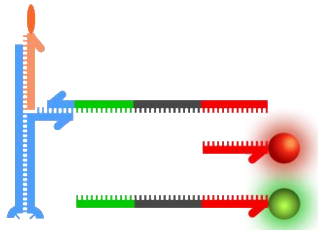
Chem, 9 (11), 3347-3364 (2023)

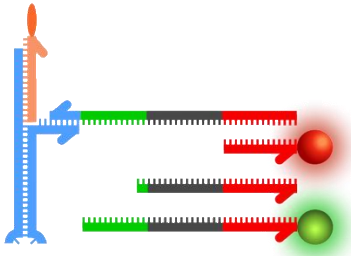
Reaction-diffusion patterning with strand displacement

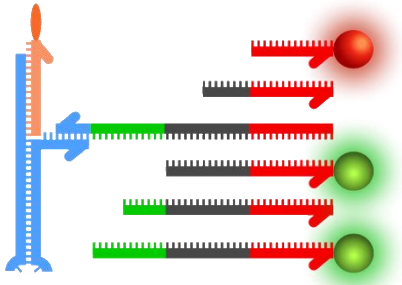


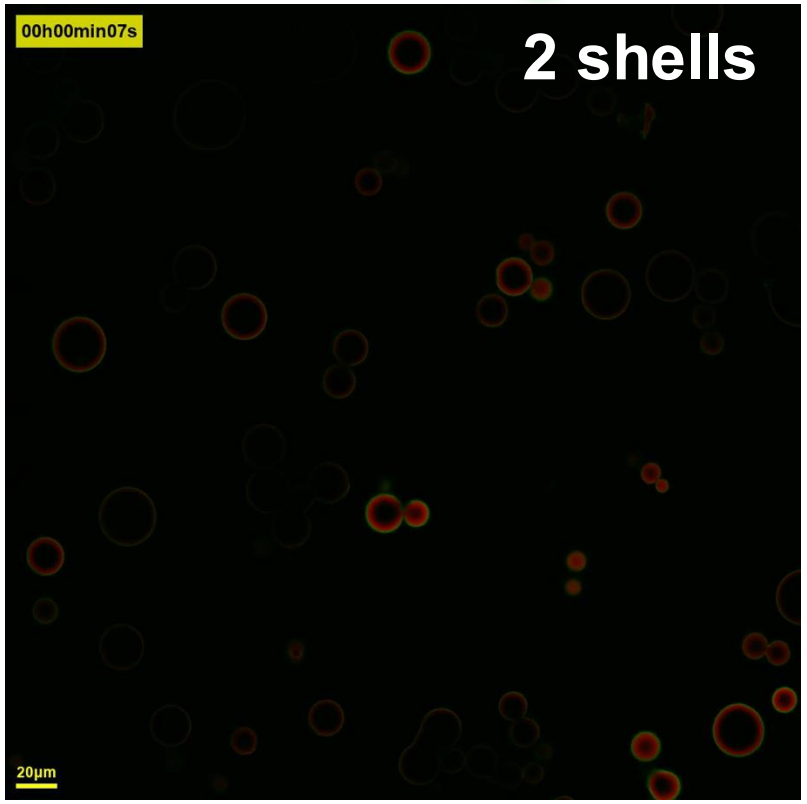
JACS, 144(38), 17468-17476 (2022)

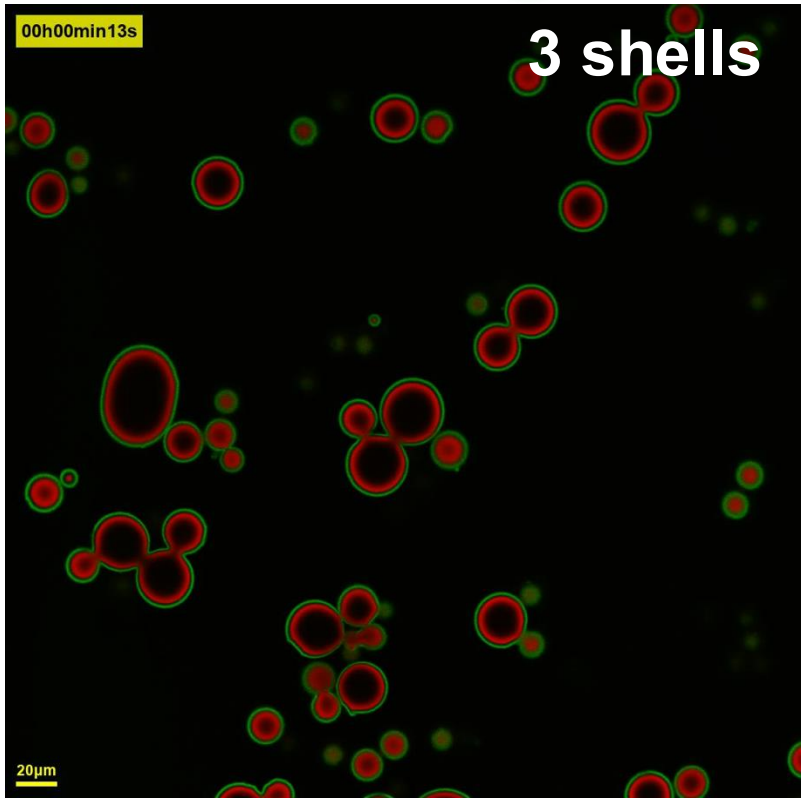
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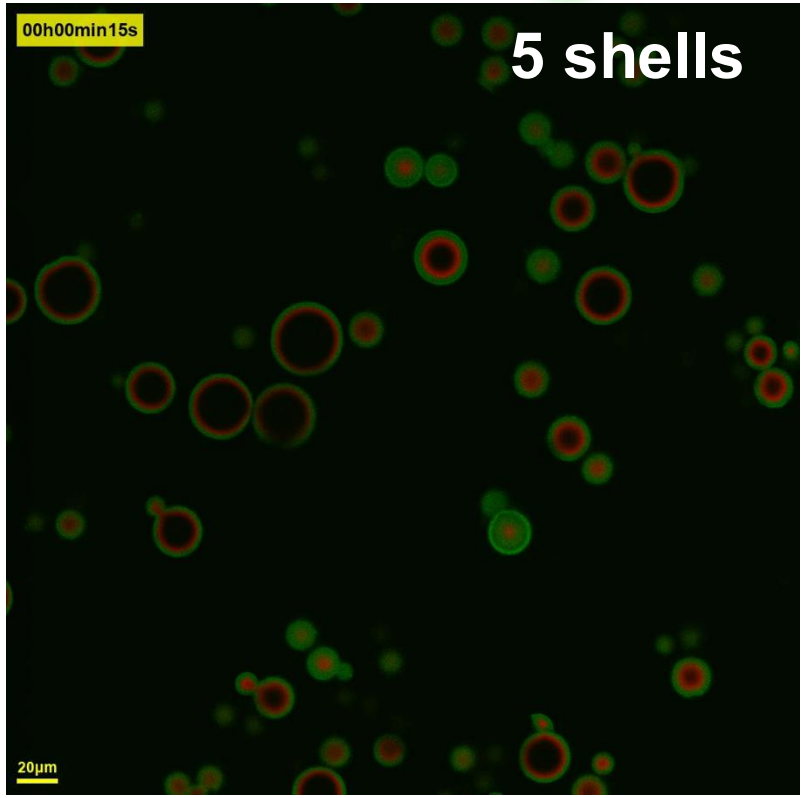






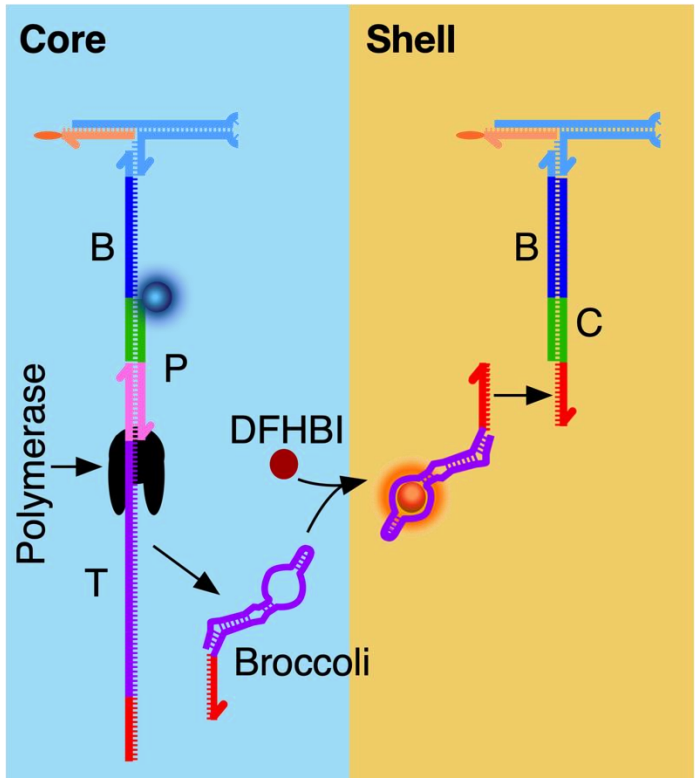


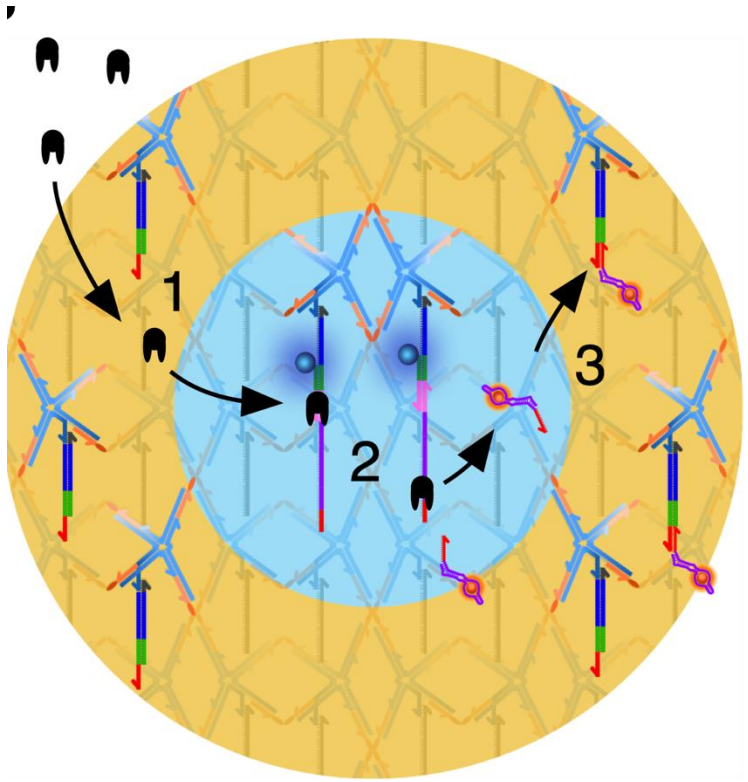


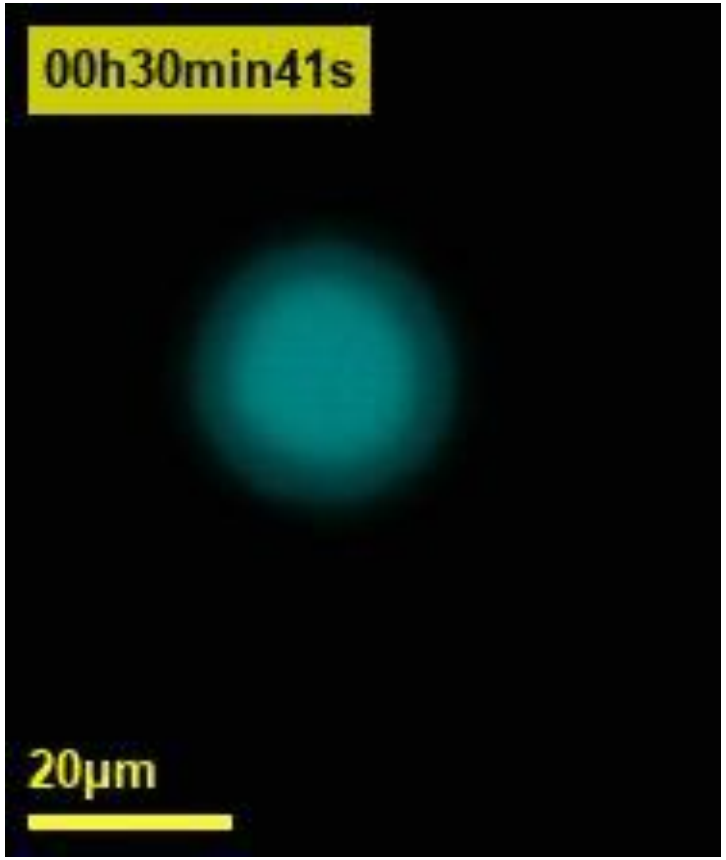


JACS, 144(38), 17468-17476 (2022)

Spatial organization of functionalities: towards synthetic organelles





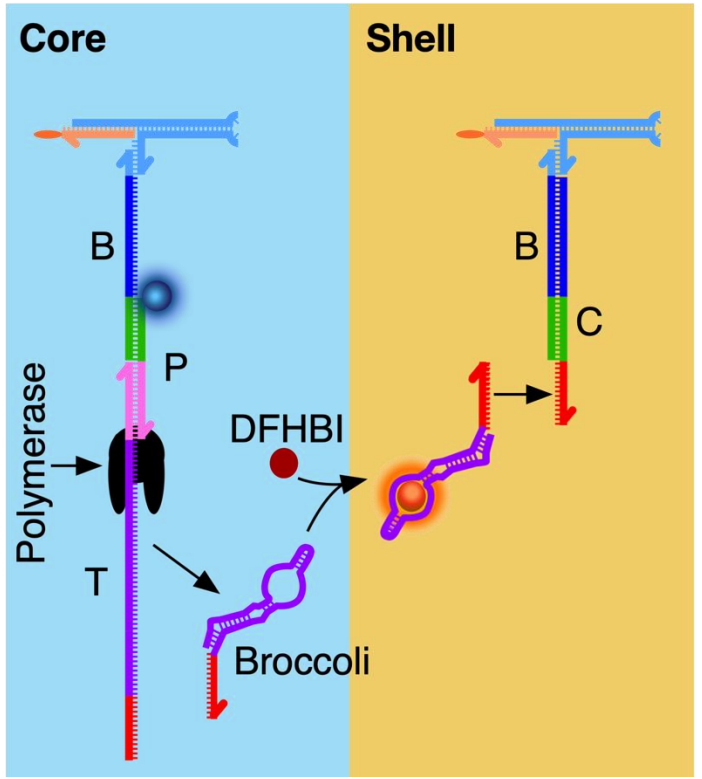


JACS, 144(38), 17468-17476 (2022)

Quiz 3

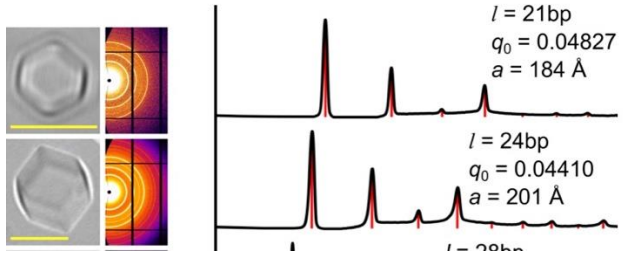
Besides RNA polymerase, what other components are needed in solution for the synthetic organelles to produce RNA?

1) NTPs, magnesium and monovalent cations
2) NTPs, magnesium, and polycations
3) dNTPs, DNA polymerase and magnesium
4) Amino acids and ribosomes



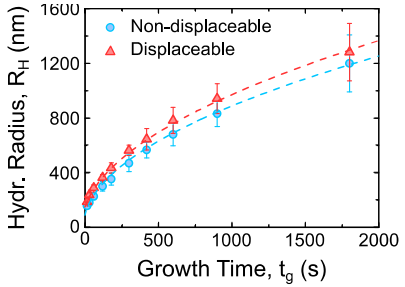
Synthetic DNA condensates

(micro)Structural control



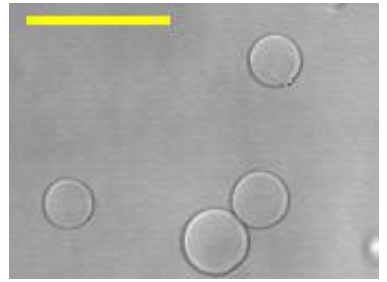
Brady et al. *JACS* 140(45)15384 (2018)

Size control



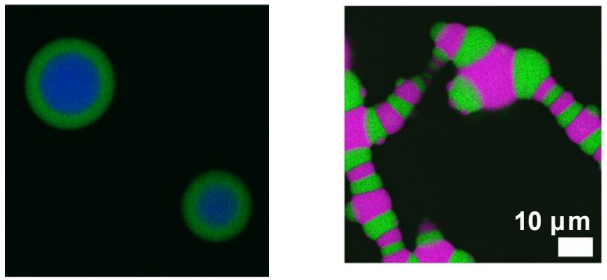
Walczak et al. *Nat. Commun.*, 12:4743 (2021)
Walczak et al. *Adv. Mater. 17, 2301562 (2023)*

Assembly/disassembly



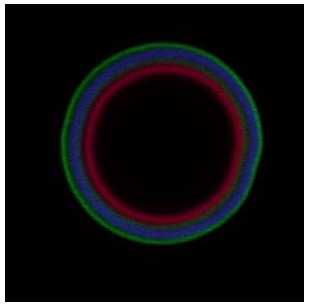
Brady et al. *JACS* 140(45)15384 (2018)
Fabrini et al. *Nano Lett.* 22 602 (2022)

Static patterning



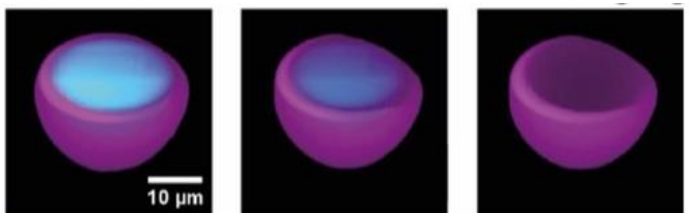
Malouf et al. *Chem*, 9, 3347 (2023)
Tanase et al. *Adv. Sci.*, e06275, (2025)

Dynamic patterning



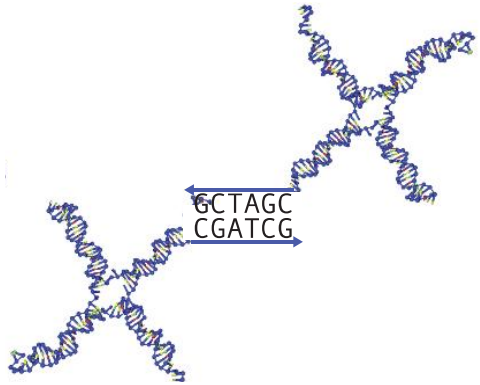
Leathers et al. *JACS*, 144, 17468, (2022)

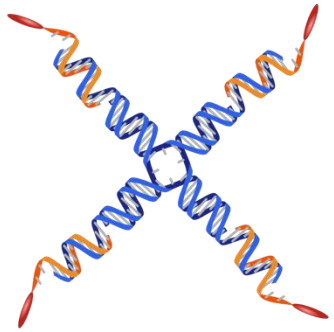
Biocatalysis

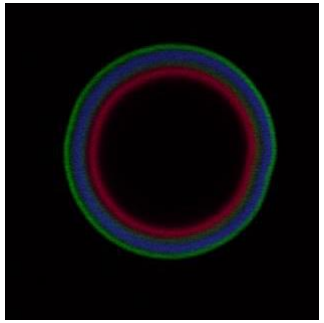


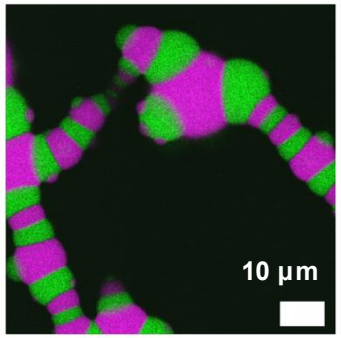
Bucci et al. *JACS*, 146, 31529 (2024)

Synthetic DNA condensates









Advantages

- Functional and stimuli-responsive
- Robust rational design
- Easy optimisation

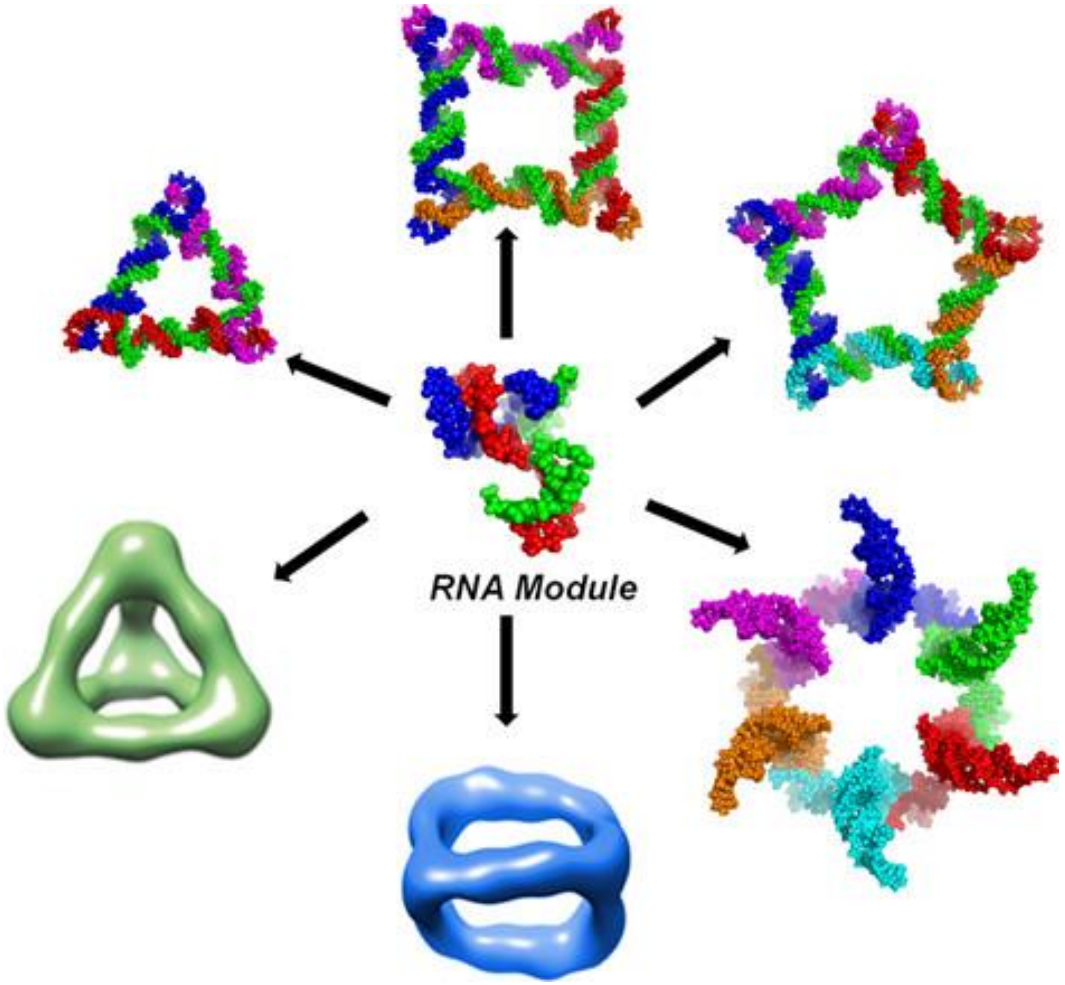
Disadvantages

- **Cannot be genetically encoded**

Lecture outline

1) Why build synthetic biomolecular condensates?
2) Engineering synthetic condensates with DNA
3) **Engineering synthetic condensates with RNA**
4) Expressing synthetic RNA condensates in cells

Can we use RNA nanotechnology?



Advantages

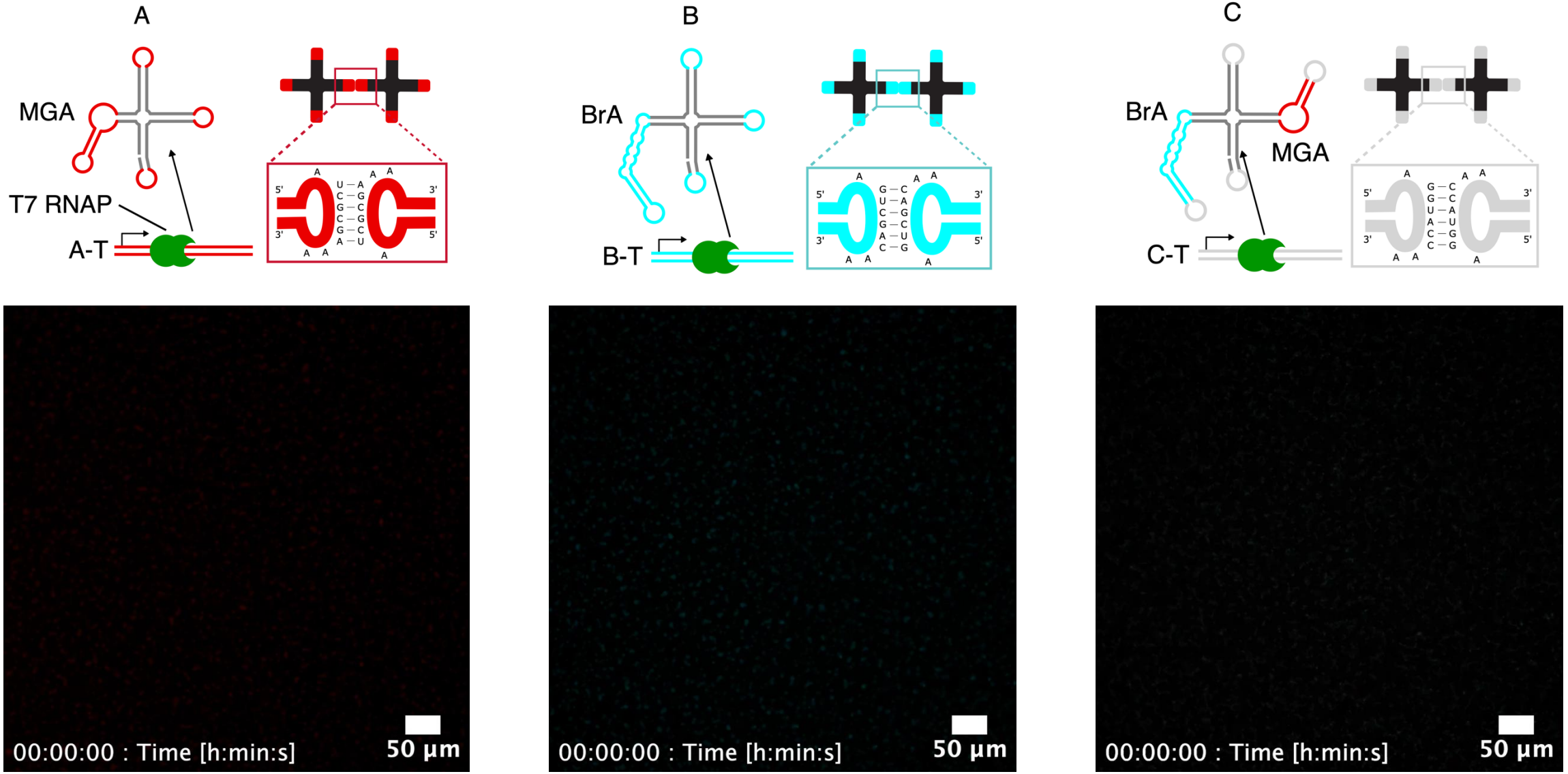
- Functional and stimuli-responsive
- Robust rational design
- Easy optimization
- Can be genetically encoded!

Disadvantages

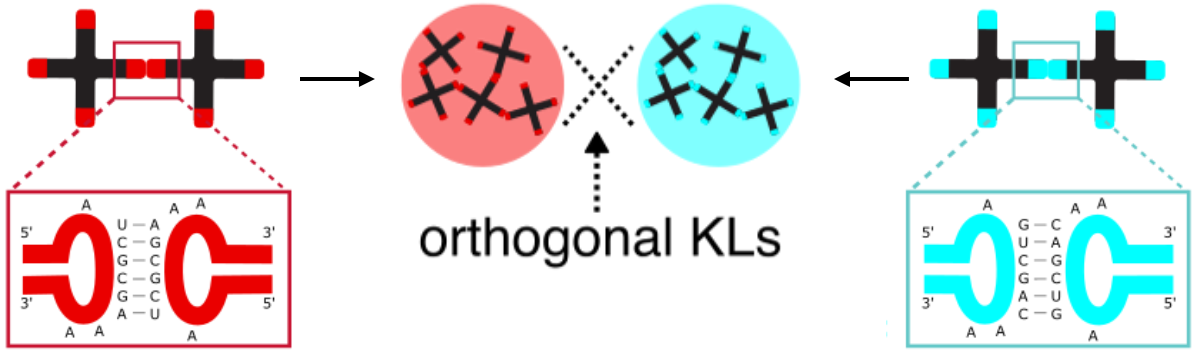
- ?

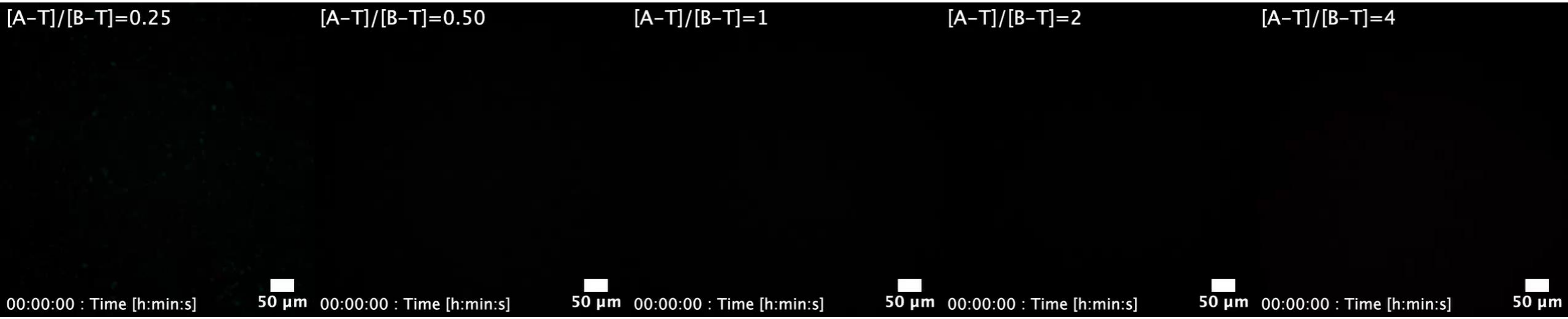
ACS Nano 2017, 11, 2, 1142-1164

Co-transcriptional assembly of RNA condensates



Orthogonal condensates





Nat. Nanotech., 19, 1665–1673 (2024)

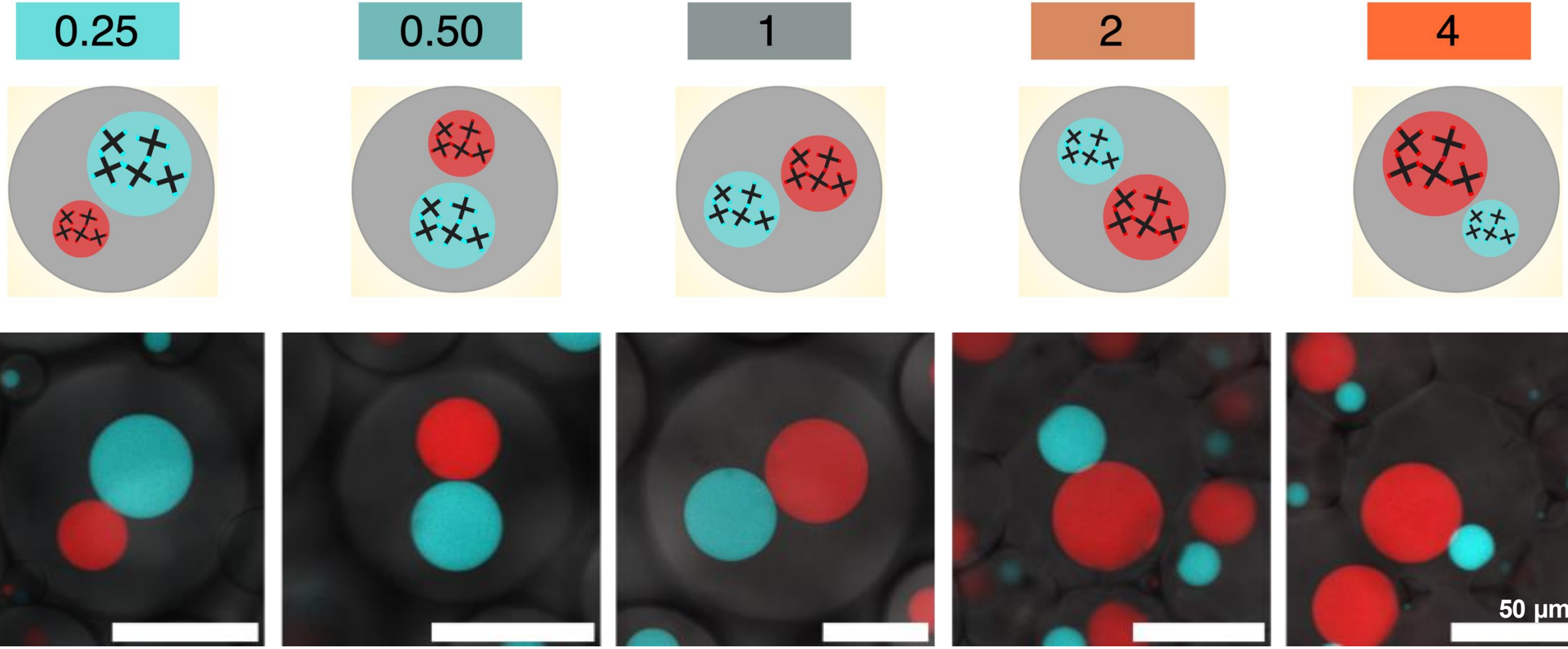
US patent application 63/588,142

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Expression in synthetic cells (w/o droplets)

[A-T] / [B-T]



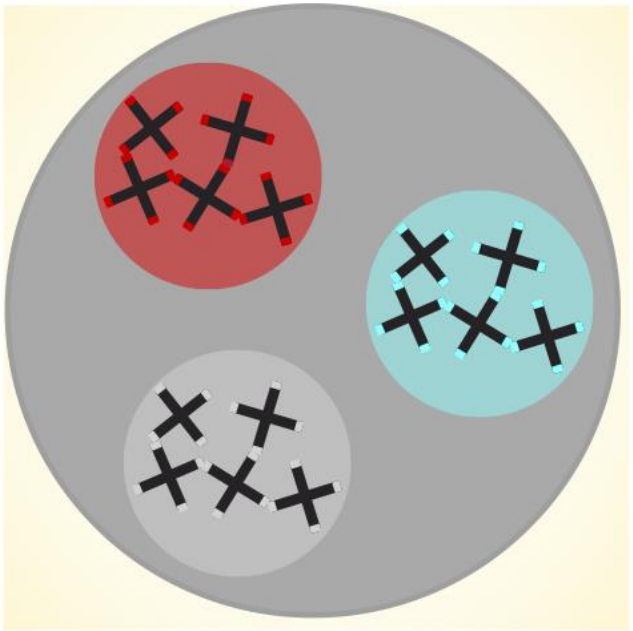
Nat. Nanotech., 19, 1665–1673 (2024)

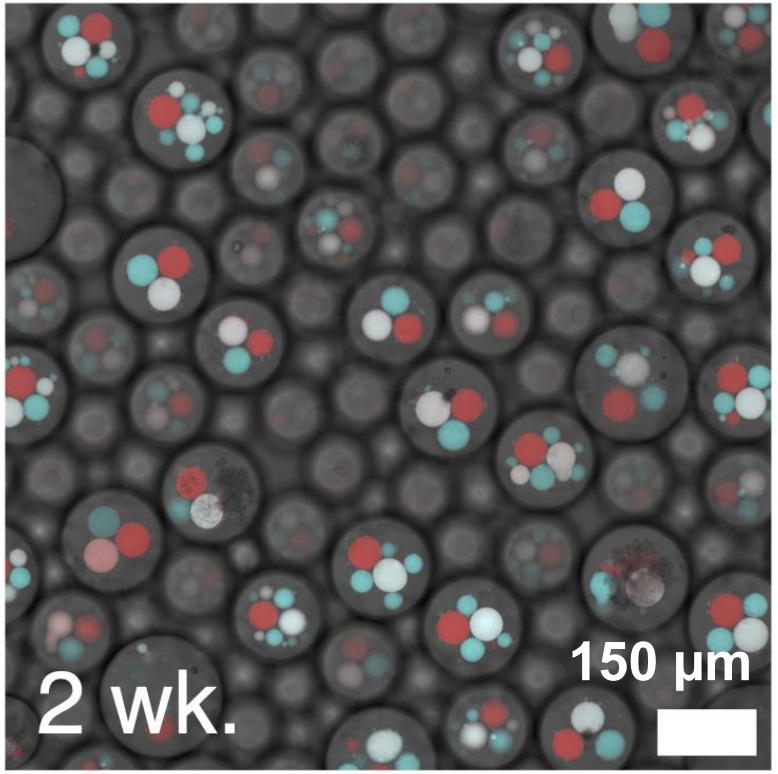
US patent application 63/588,142

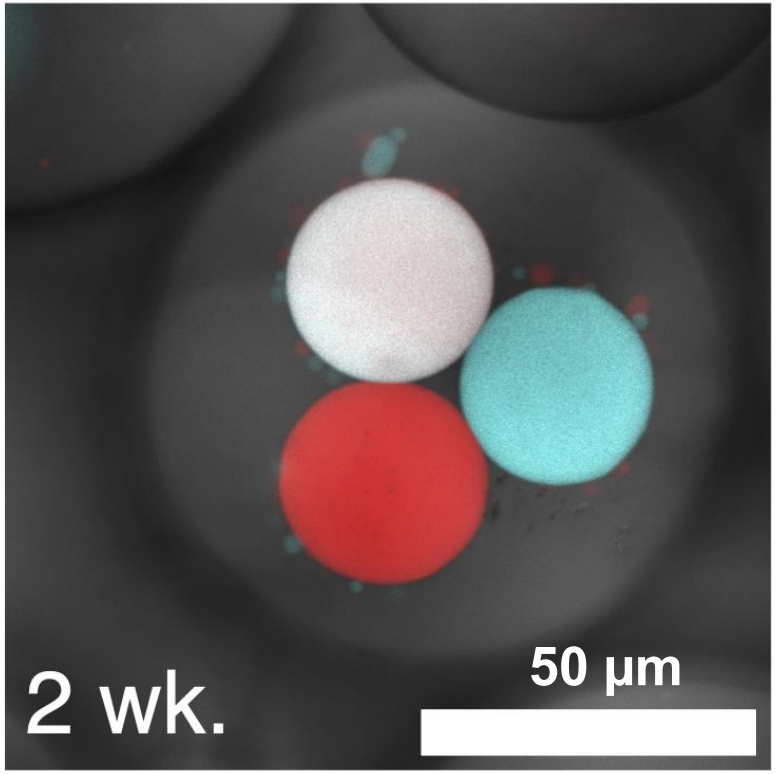
Di Michele Lab
University of Cambridge

University of Cambridge

Expression in synthetic cells (w/o droplets)



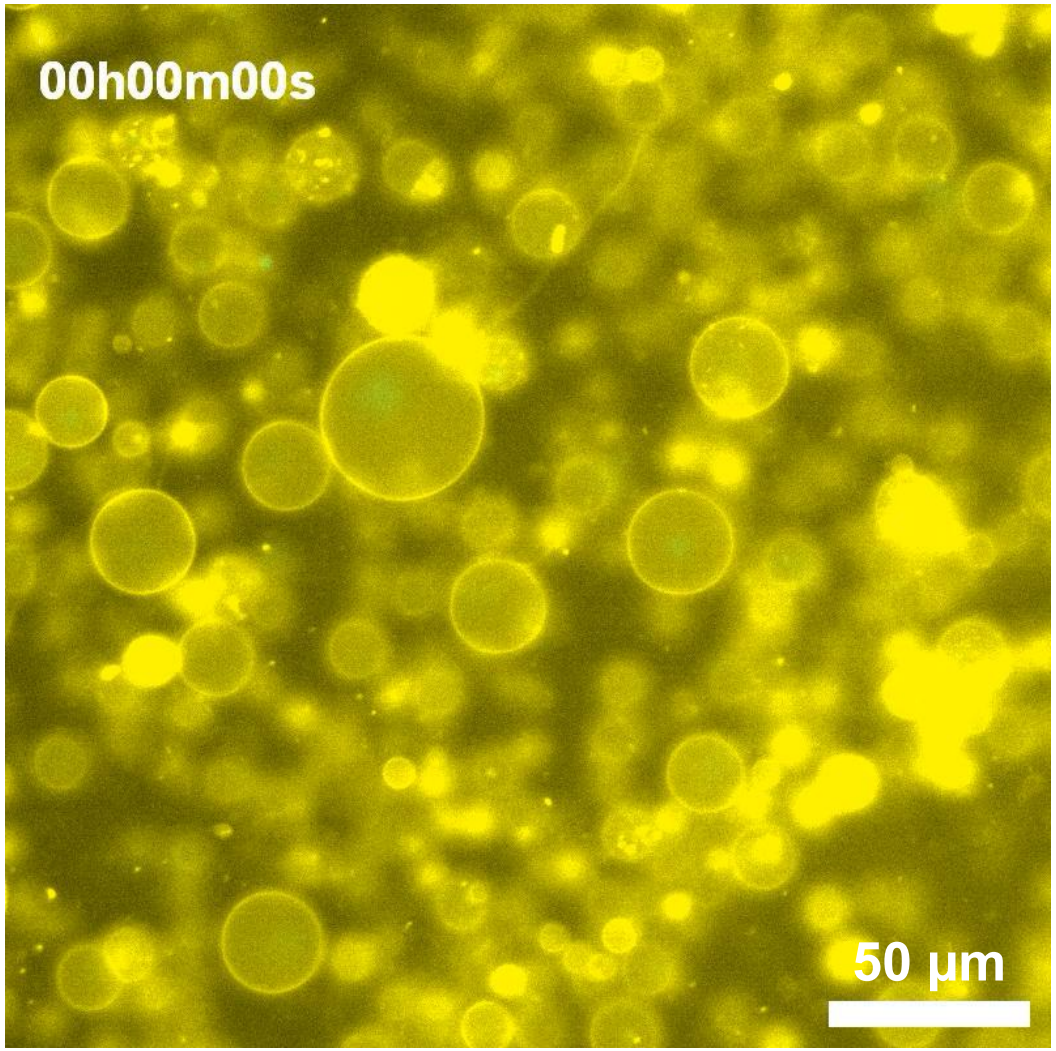


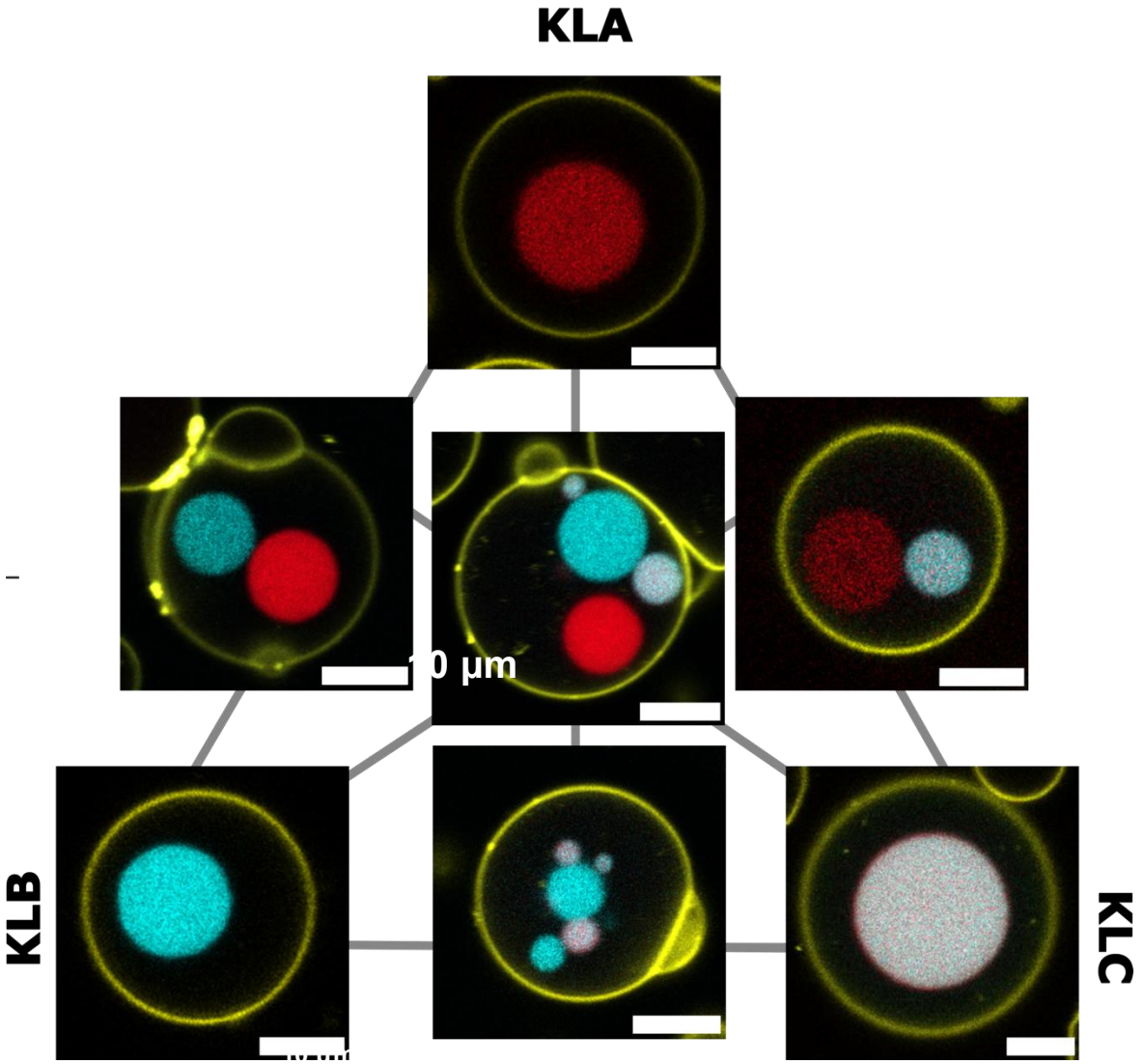


Nat. Nanotech., 19, 1665–1673 (2024)

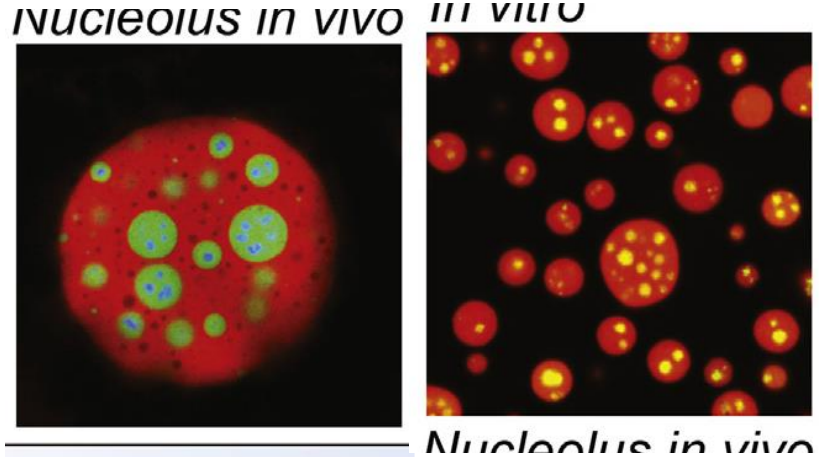
US patent application 63/588,142

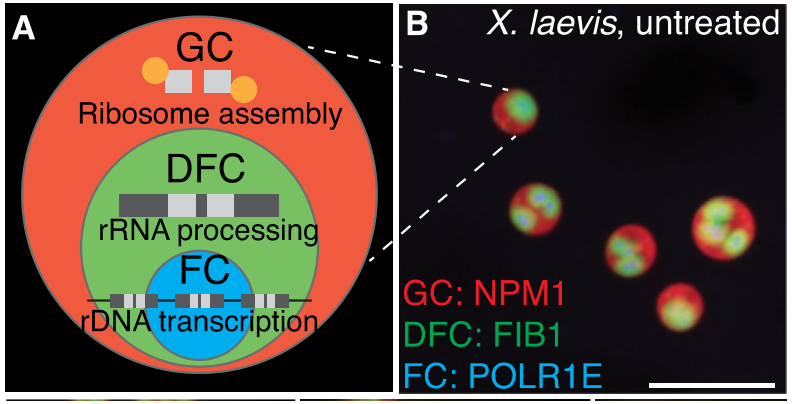
Expression in synthetic cells



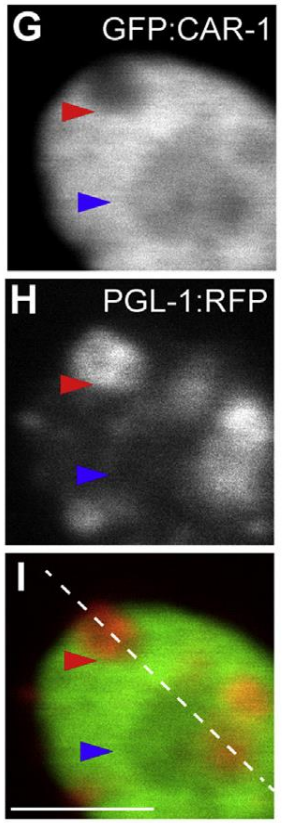


Multiphase condensates



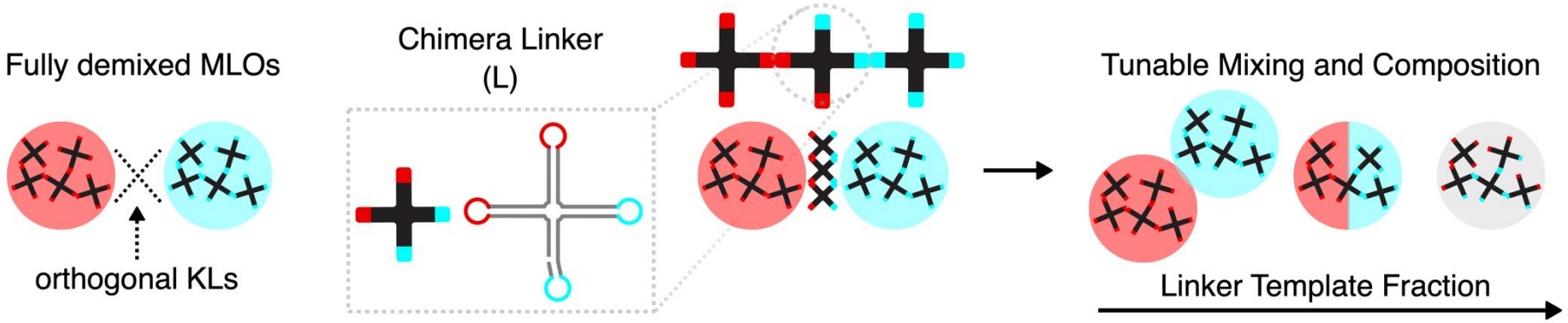


Feric *et al.*, Cell, 165, 1686, 2016



Dev. Cell 27, 161, 28, 2013

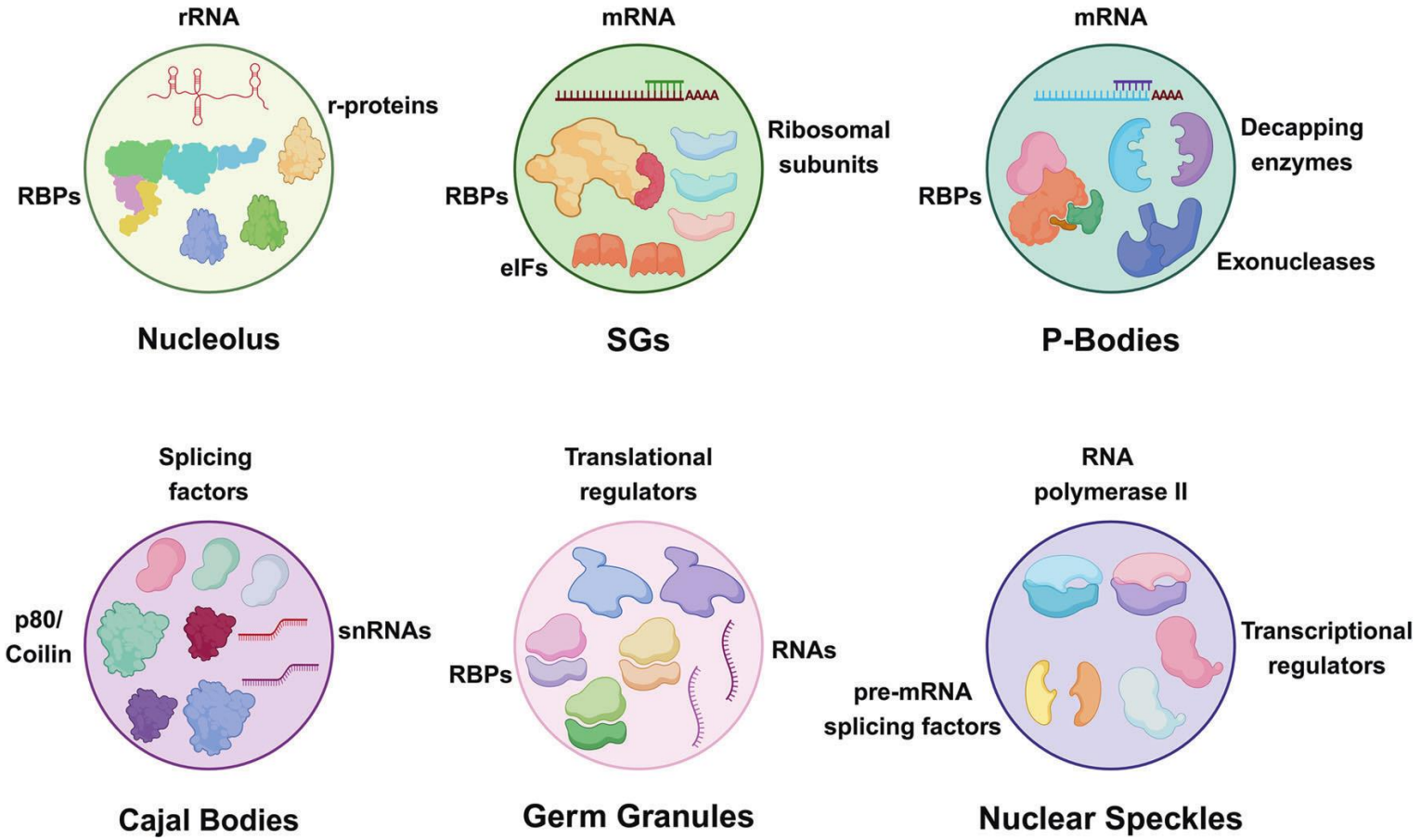
Multiphase condensates



Nat. Nanotech., 19, 1665–1673 (2024)

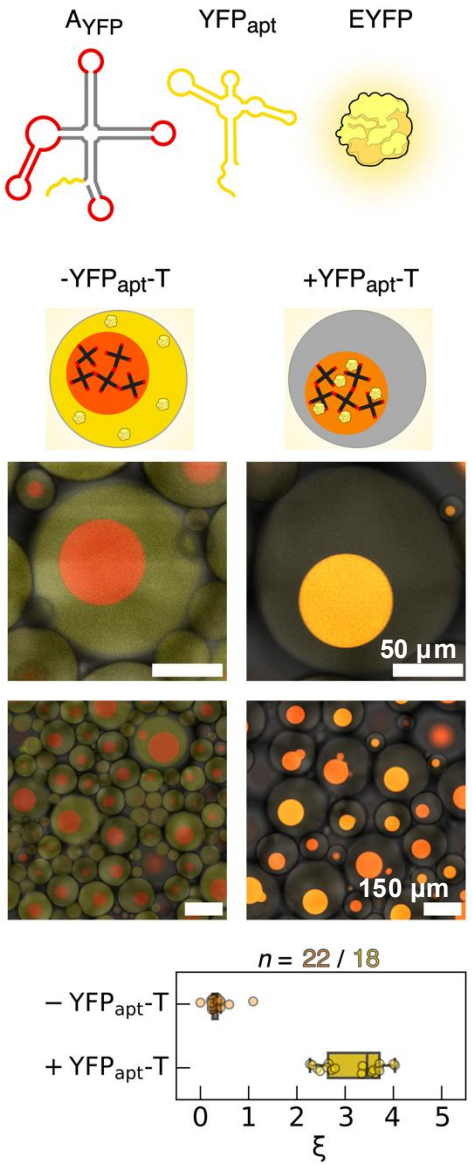
US patent application 63/588,142

Selective capture of proteins



Signal Transduction and Targeted Therapy 9:305 (2024)

Selective capture of proteins



Nat. Nanotech., 19, 1665–1673 (2024)

US patent application 63/588,142

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Quiz 4

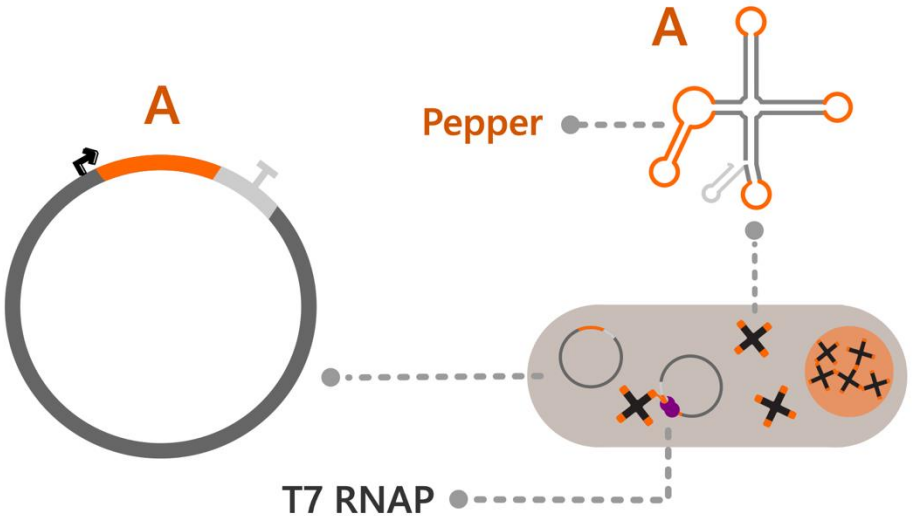
Which of the following techniques are likely to have been involved in the selection of the YFP-binding RNA aptamers?

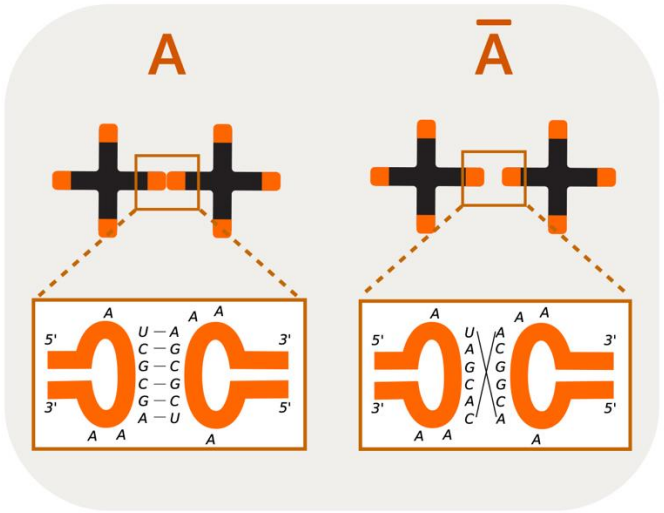
1) RT-PCR
2) In vitro transcription
3) Sequencing
4) Some of the above
5) All of the above

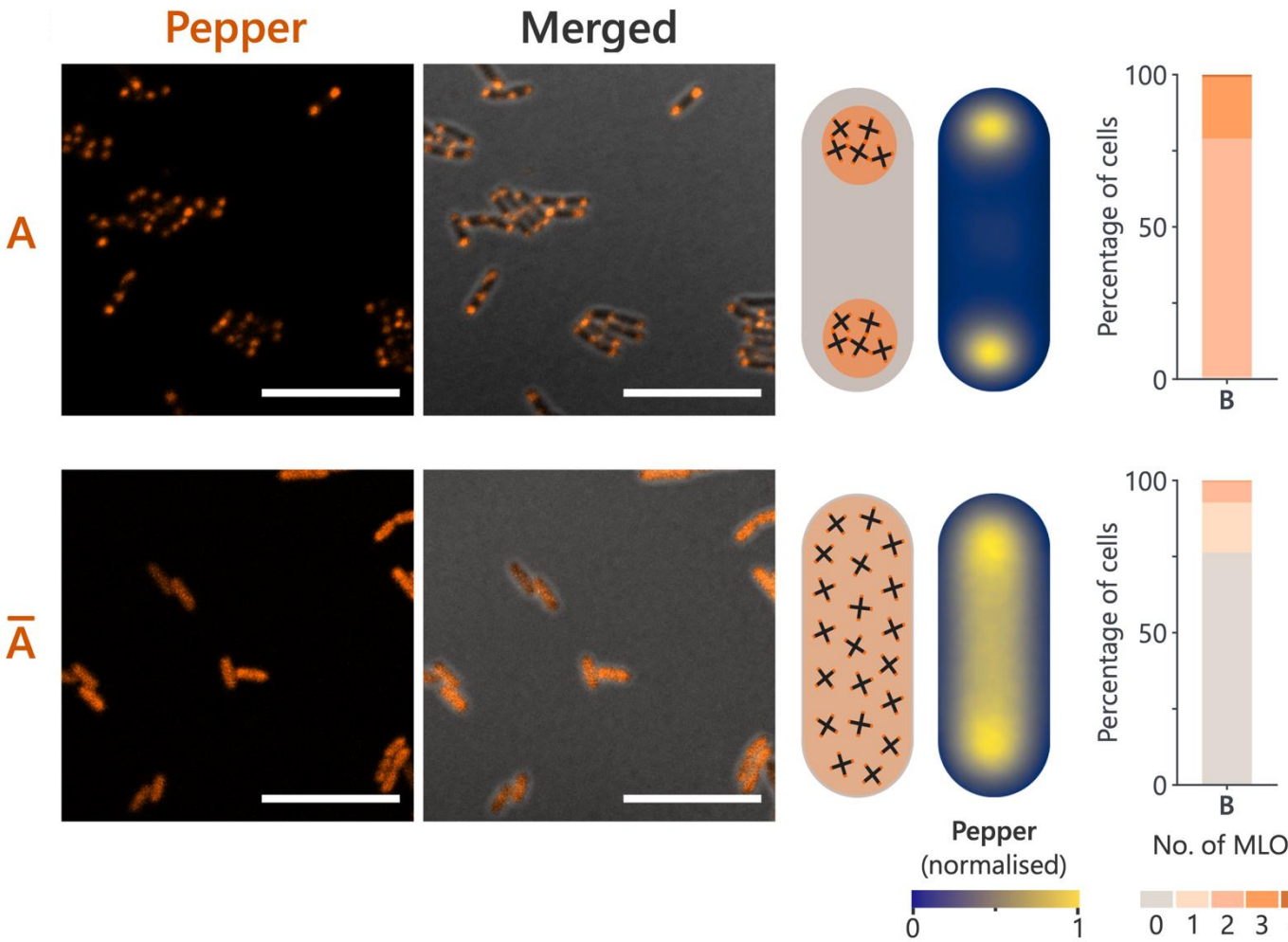
Lecture outline

1) Why build synthetic biomolecular condensates?
2) Engineering synthetic condensates with DNA
3) Engineering synthetic condensates with RNA
4) **Expressing synthetic RNA condensates in cells**

Expression of RNA Membrane-Less Organelles in *E. coli*

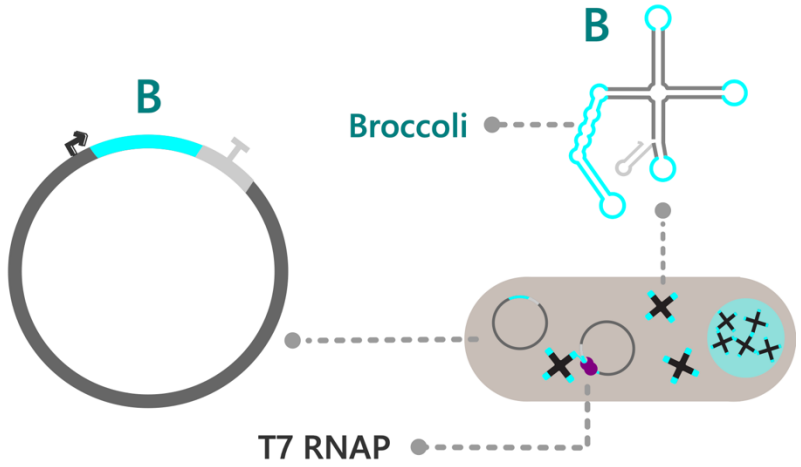


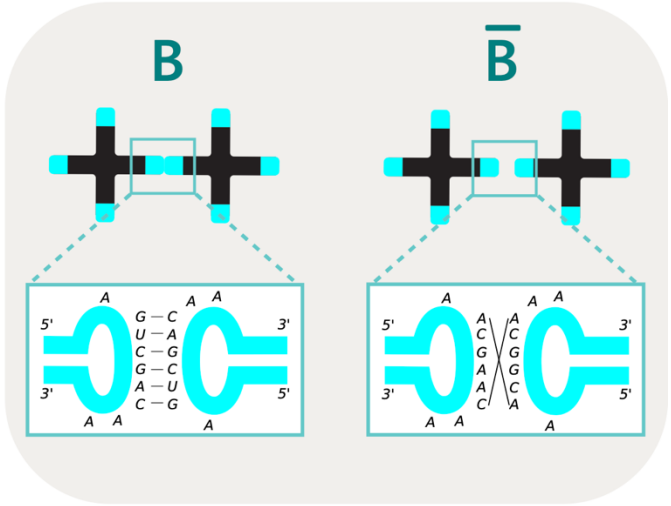


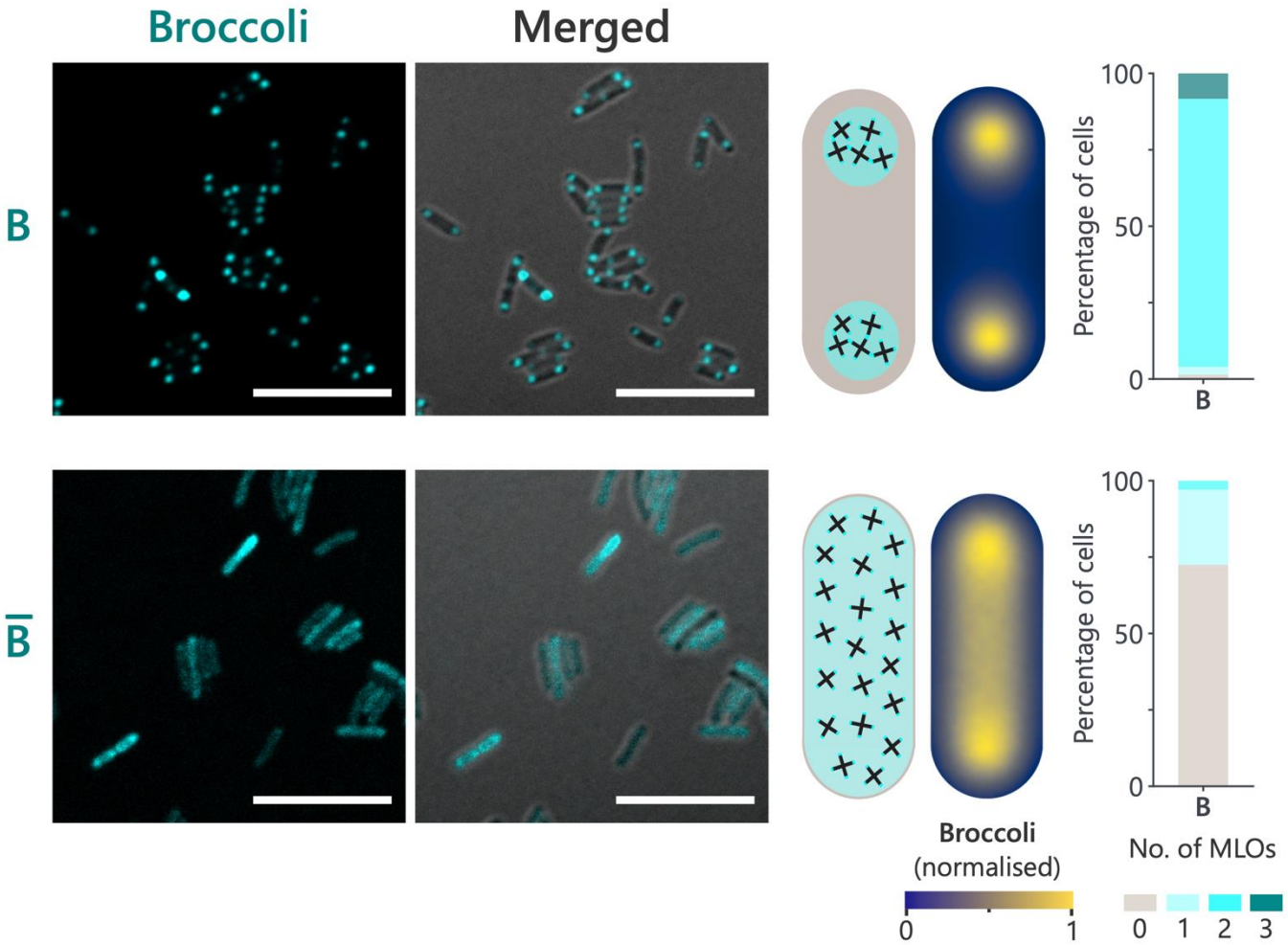


B. Ng *et al.*, submitted – UK patent application 2510955.4

Expression of RNA Membrane-Less Organelles in *E. coli*



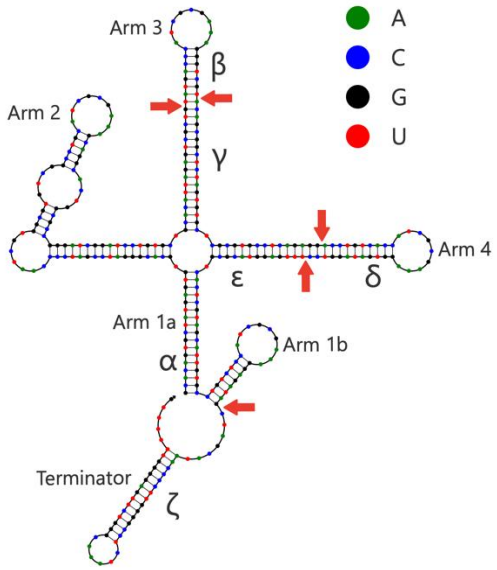




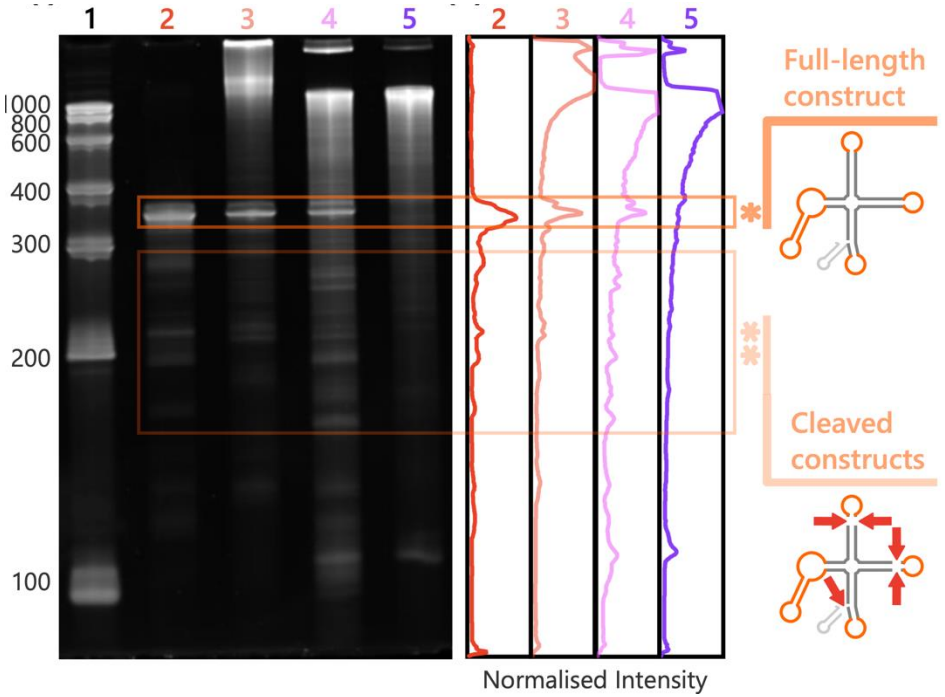
B. Ng *et al.*, submitted – UK patent application 2510955.4

Enzymatic processing: Nanostar A

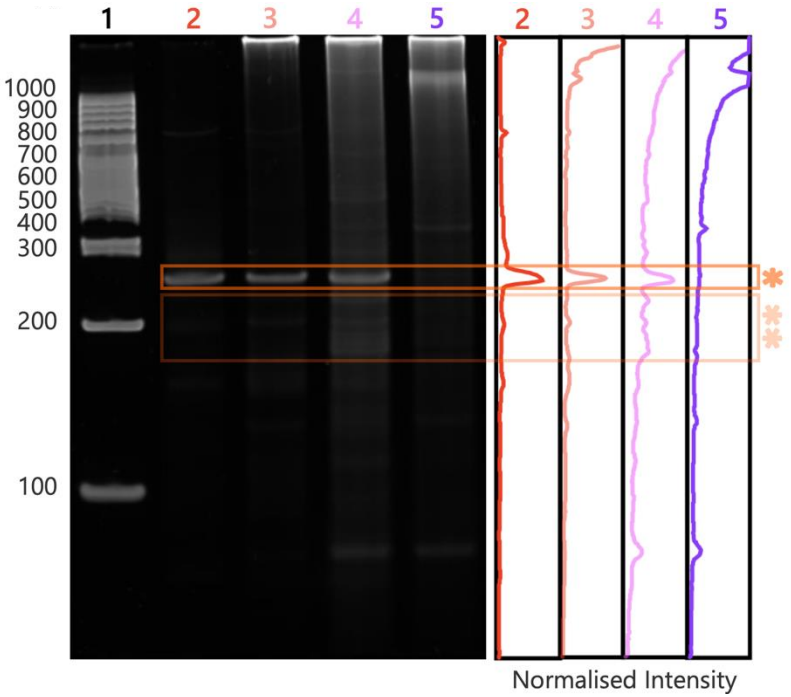
From sequencing



Denaturing PAGE



Native PAGE



1) Ladder; 2) In vitro transcription from linear DNA; 3) In vitro transcription from plasmid; 4) Induced *E. coli*; 5) Non-induced *E. coli*

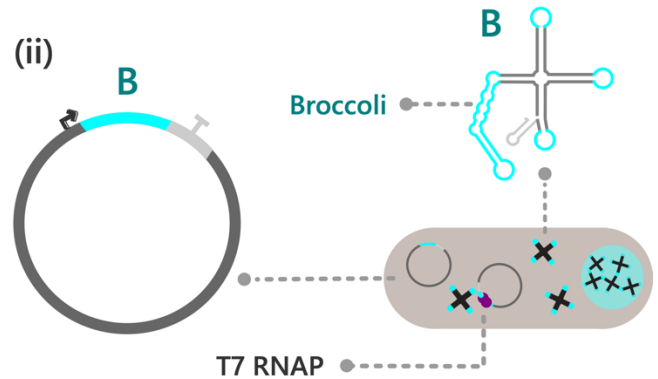
Quiz 5

Assuming that:

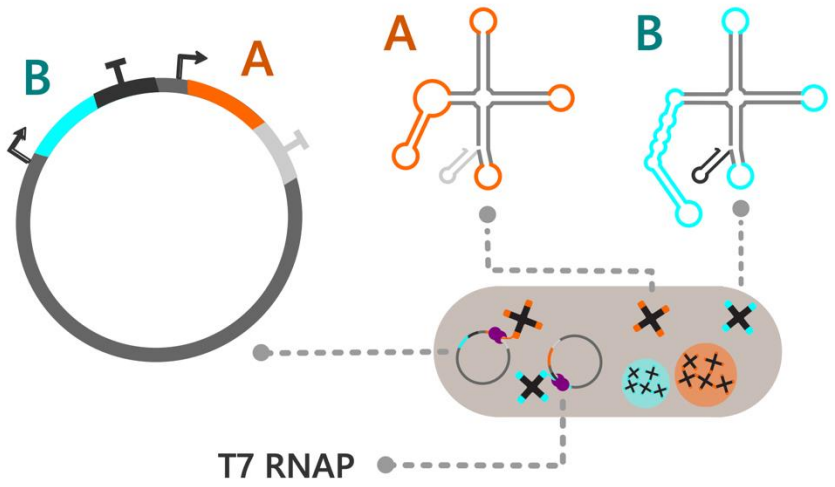
- The transcription rate of T7 RNA polymerase is $k = 250\ \mathrm{nt\,s^{-1}}$
- The RNA nanostars are $250\ \mathrm{nt}$ in length
- There are $\sim 30{,}000$ nanostars per *E. coli* cell
- The copy number of the plasmid containing the nanostar template is 25

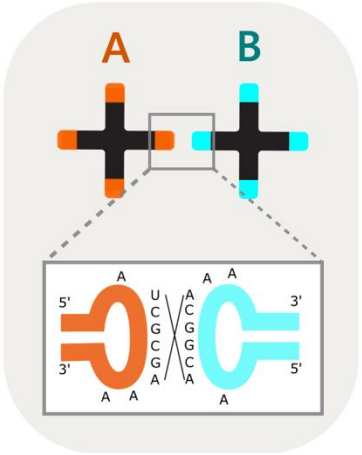
What is the lifetime of the RNA nanostars in *E. coli* ?

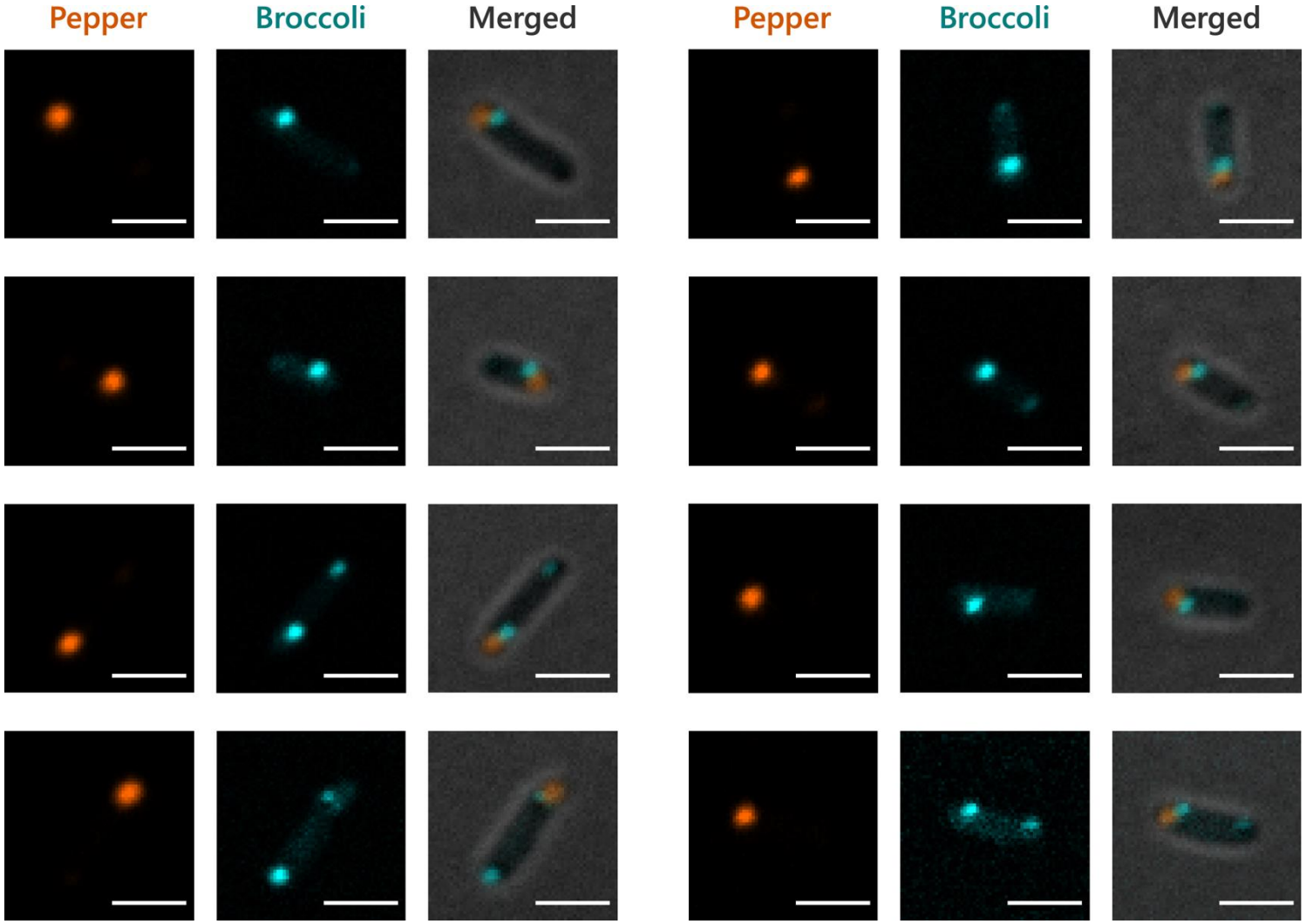
1) 2 minutes
2) 10 minutes
3) 20 minutes
4) 100 minutes



Orthogonal Membrane-Less Organelles

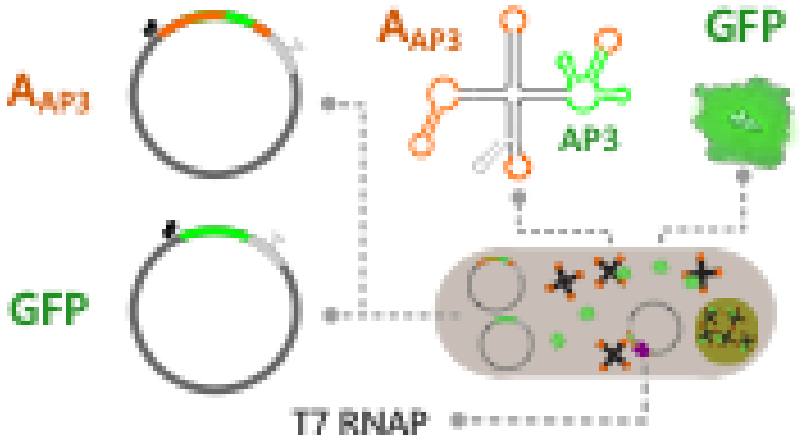


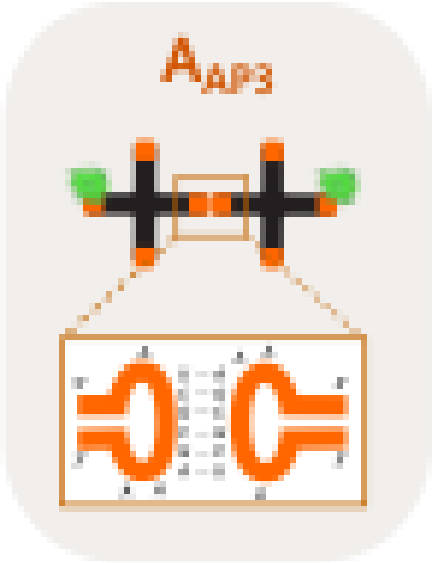


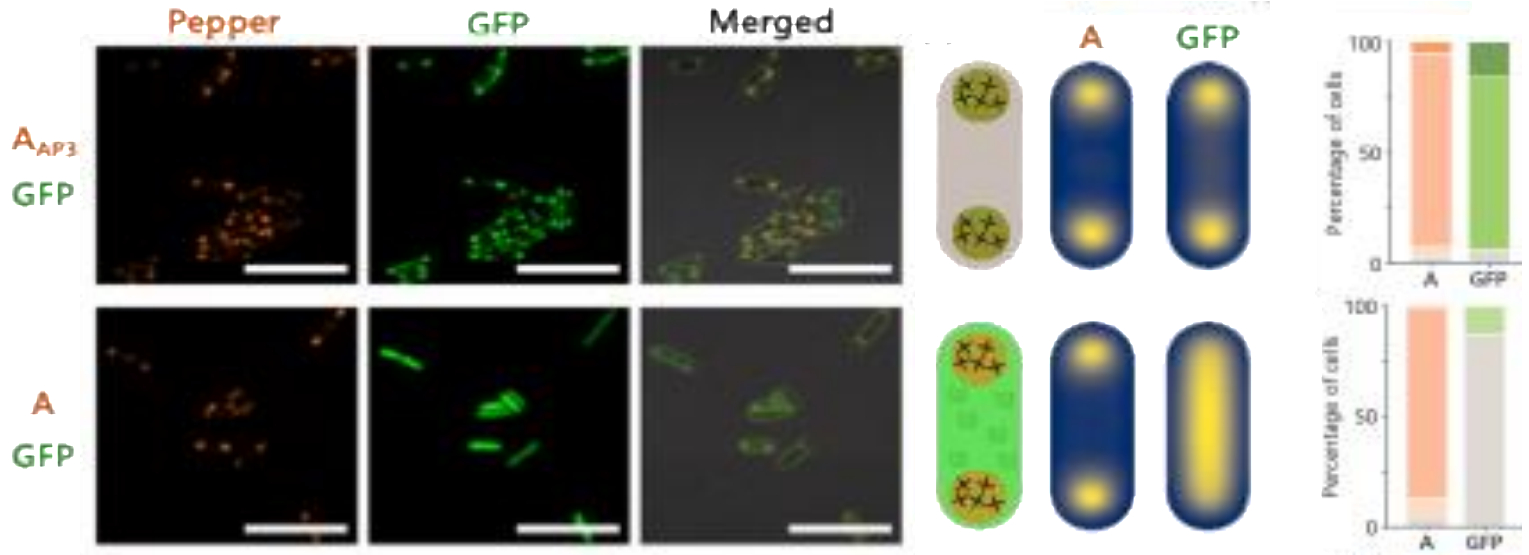


B. Ng *et al.*, submitted – UK patent application 2510955.4

Selective protein capture







B. Ng *et al.*, submitted – UK patent application 2510955.4

That's all

Thanks for your attention!



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